



MATHS in the Asia Pacific

Paul Steenson



Quality ready-to-use resources



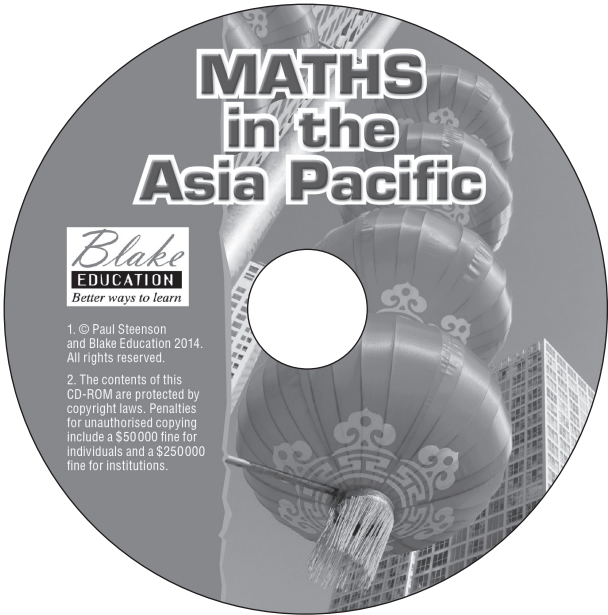
**Made for the
Australian Curriculum**



Maths in the Asia Pacific

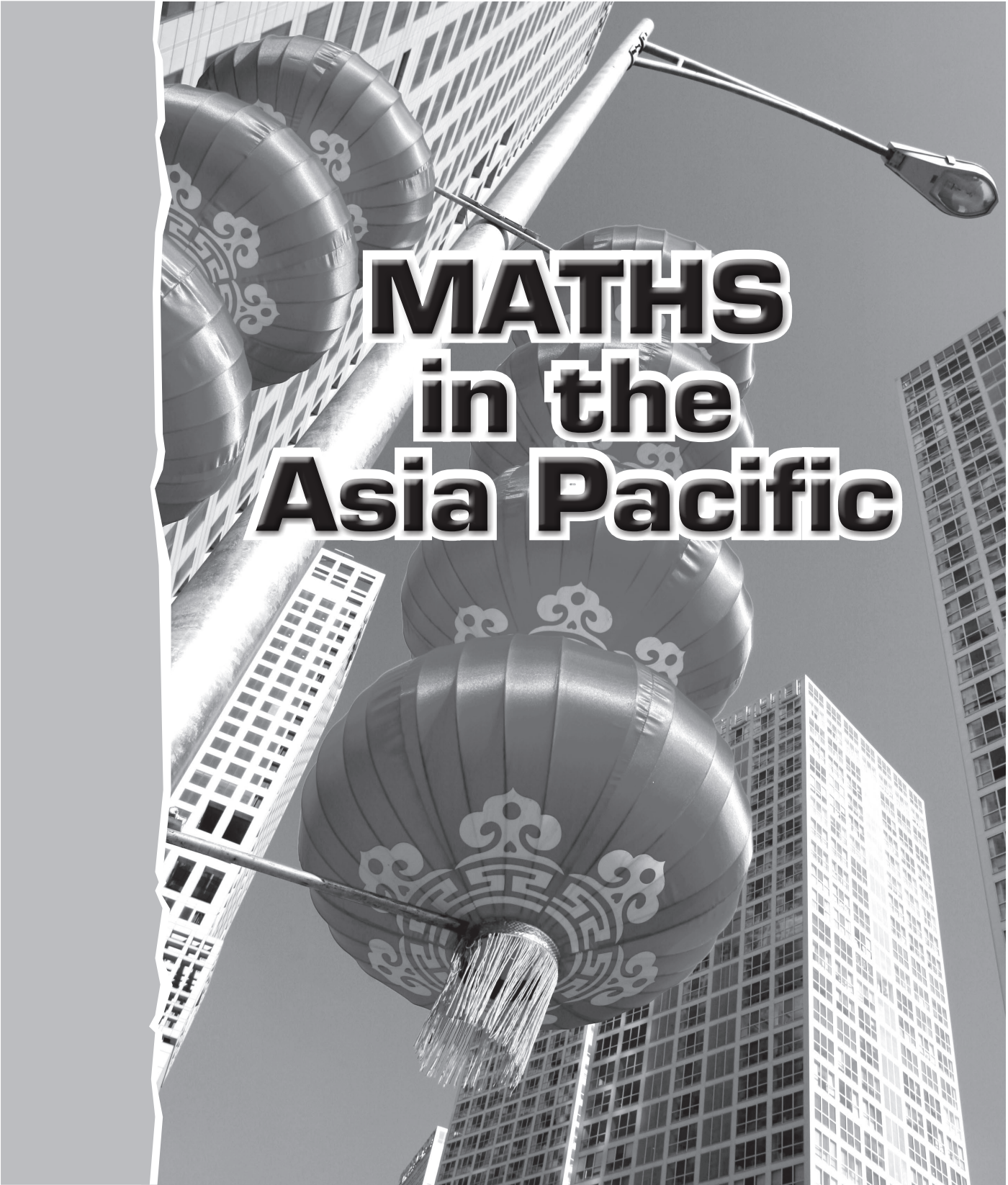
Features

- ✓ Quality ready-to-use resources
- ✓ Activities designed to develop core mathematical skills
- ✓ Informative topics that will engage secondary school students
- ✓ Aligned to the Australian Curriculum
- ✓ Answers provided for easy and efficient marking
- ✓ CD-ROM containing the book in PDF format, ideal for projecting onto the interactive whiteboard



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Australian curriculum alignment	Year	Units
ACMNA280 Compare, order, add and subtract integers	7	1
ACMNA155 Express one quantity as a fraction of another, with and without the use of digital technologies	7	2
ACMNA158 Find percentages of quantities and express one quantity as a percentage of another, with and without digital technologies	7	2, 3, 7, 11, 18, 19
ACMNA173 Recognise and solve problems involving simple ratios	7	6, 8
ACMNA180 Investigate, interpret and analyse graphs from authentic data	7	16
ACMNA188 Solve a range of problems involving rates and ratios, with and without digital technologies	8	4, 5, 6, 7, 8
ACMMG159 Establish the formulas for areas of rectangles, triangles and parallelograms and use these in problem solving	7	10
ACMMG181 Describe translations, reflections in an axis, and rotations of multiples of 90° on the Cartesian plane using coordinates. Identify line and rotational symmetries	7	9
ACMMG196 Find perimeters and areas of parallelograms, trapeziums, rhombuses and kites	8	11
ACMMG198 Develop the formulas for volumes of rectangular and triangular prisms and prisms in general. Use formulas to solve problems involving volume	8	11
ACMMG199 Solve problems involving duration, including using 12- and 24-hour time within a single time zone	8	12, 13
ACMSP169 Identify and investigate issues involving numerical data collected from primary and secondary sources	7	7, 8, 14, 15, 16
ACMSP171 Calculate mean, median, mode and range for sets of data. Interpret these statistics in the context of data	7	17, 18, 19
ACMSP228 Identify everyday questions and issues involving at least one numerical and at least one categorical variable, and collect data directly from secondary sources	9	17
ACMSP253 Evaluate statistical reports in the media and other places by linking claims to displays, statistics and representative data	10	18, 19



MATHS **in the** **Asia Pacific**

Paul Steenson

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Paul Steenson has more than thirty years' experience teaching Mathematics in secondary schools in Australia and has been Head of Department in a variety of schools in New South Wales. He has also worked as a test development officer and been an assessor of Mathematics papers for the NSW HSC examination. Paul served as a member of the NSW Mathematics Syllabus Committee and has been a regular presenter of curriculum workshops and teacher development activities. He has also been a co-author of textbooks and written a number of titles in the *Instant Lessons* series. Paul has travelled widely and has made multiple visits to many of the countries mentioned in this book.

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Maths in the Asia Pacific is a collection of photocopiable student work sheets that feature the application of mathematical knowledge and skills in engaging, real-world contexts. Travel, art, history, animals and geography are some of the topics used to provide fresh approaches for teaching Australian Curriculum content. The activities will help students recognise how mathematical skills can be used in daily life.

The work sheets are designed to be used in the classroom or for homework assignments across years 7 to 10. All work sheets are included in the PDF version of the book on the enclosed CD-ROM.

Links to the Australian Curriculum can be found in the alignment grid on the inside front cover of this text. Work sheets 20 (Chinese numerals) and 21 (The abacus) are included as extension activities and have not been linked to the curriculum in the alignment grid.

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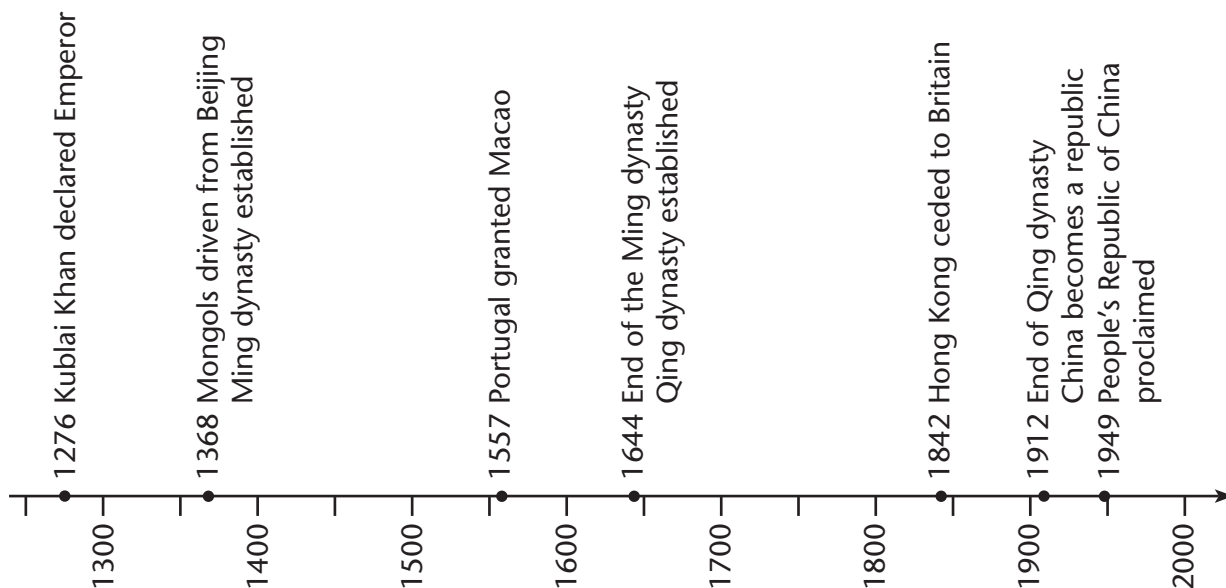
Time lines in Asian history



1

Time lines are examples of number lines. The dates of events are arranged in ascending order, left to right. A suitable scale should be used to indicate the correct gaps between the dates.

This time line shows the dates of key events in the history of **China**.



- 1 In which year was
 - a Kublai Khan declared Emperor of China?
 - b the Qing dynasty established?
 - c Hong Kong ceded to Britain?
 - d the People's Republic of China proclaimed?
- 2 Which event happened earlier – A or B?
 - A Hong Kong was ceded to Britain.
 - B Portugal was granted Macao
(a small peninsula in southern China).
- 3
 - a In which year was the Ming dynasty established?
 - b In which year did the Ming dynasty end?
 - c How many years did the Ming dynasty last?
- 4 How many years did the Qing dynasty last?
- 5 How many years after Kublai Khan was declared Emperor
 - a was the Ming dynasty established?
 - b was the Qing dynasty established?
 - c did China become a republic?
- 6 Hong Kong was handed back to China in 1997.
How many years did British rule last in Hong Kong?

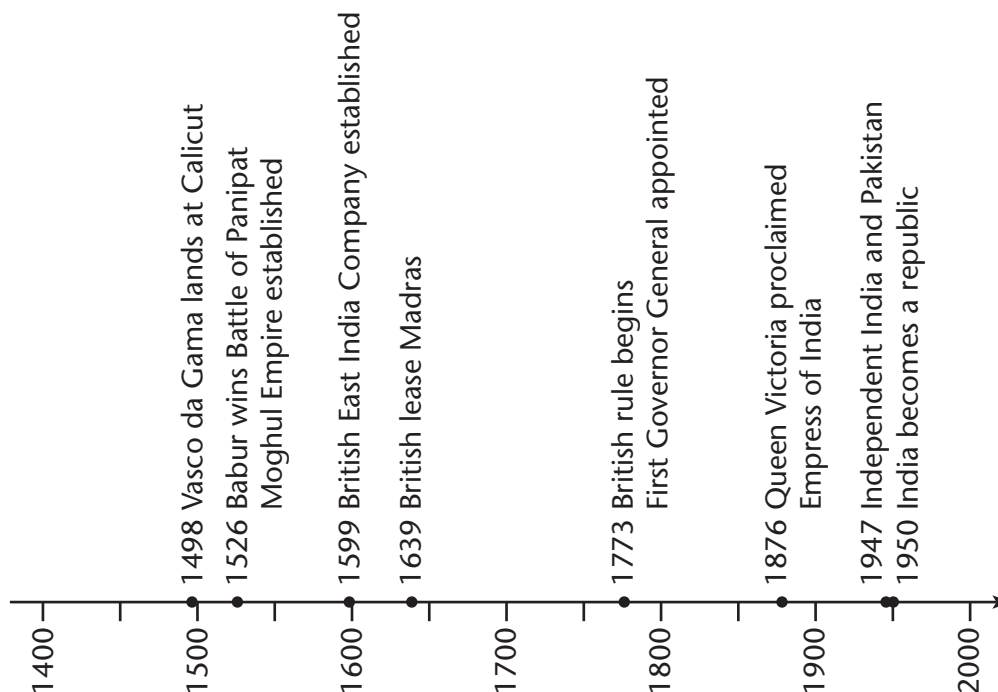


Time lines in Asian history



1

This time line shows the dates of key events in the history of **India**.



- 7 In which year did
 - a Babur win the Battle of Panipat?
 - b British rule of India begin?
 - c India become a republic?
- 8 Which event happened earlier – A or B?
 - A Babur won the Battle of Panipat.
 - B The British East India company was established.
- 9 How many years after the beginning of British rule in India
 - a was Queen Victoria proclaimed Empress of India?
 - b did India become a republic?
- 10 Draw a 500 year time line 10 cm long.

Mark the end points with the years 1500 and 2000.

Mark the years 1600, 1700, 1800 and 1900 at intervals 2 cm apart.

Mark the following dates of key events in the history of **Japan** on your time line.

 - 1603 Tokugawa Ieyasu becomes Shogun
 - 1635 Closed door policy restricts entry by foreigners
 - 1707 Mt Fuji erupts
 - 1867 The Meiji restoration; rule of the shoguns ends
 - 1923 Earthquake devastates Tokyo
 - 1945 Japan surrenders at the end of World War II
 - 1956 Japan becomes a member of the United Nations





Papua New Guinea is home to a large number of species of birds.

The **bird-of-paradise** is universally identified with the island of New Guinea. An adult bird-of-paradise is depicted on the national flag of Papua New Guinea.

There are 44 species of birds-of-paradise worldwide and 35 species are found in Papua New Guinea. A few species are also found in surrounding islands and north-eastern Australia.

The smallest bird-of-paradise in size is the **King Bird-of-paradise** which on average has a mass of 50 g and is 15 cm long.

The **Curl-crested Manucode** is 44 cm long and has a mass of 430 g.

The male **Black Sicklebill**, with its long tail, is the longest bird-of-paradise species at 110 cm.



- 1 What fraction of the 44 species of birds-of-paradise are found in Papua New Guinea?
- 2 Express the average mass of a King Bird-of-paradise as a fraction of the average mass of a Curl-crested Manucode. Simplify your answer.
- 3 Express the length of a Curl-crested Manucode as a fraction of the length of a male Black Sicklebill. Simplify your answer.
- 4 Express the length of a King Bird-of-paradise as a percentage of the length of a male Black Sicklebill. Give your answer correct to 1 decimal place.

There are many other types of birds found in New Guinea.

The **Papuan Eagle** (also known as the Kapul Eagle) has an average length of 83 cm. Its wingspan measures 157 cm and its average mass is 2 000 g.

The **Palm Cockatoo** has an average mass of 1 060 g.

- 5 a Express the average mass of a Palm Cockatoo as a fraction of the average mass of a Papuan Eagle. Simplify your answer.
b Write this answer as a percentage.
- 6 The Palm Cockatoo is about $\frac{7}{10}$ of the length of a Papuan Eagle.

What is the average length of a Palm Cockatoo?
Answer to the nearest centimetre.



The **Blue-faced Honeyeater** has distinctive plumage, with olive upperparts, white underparts and a black head and throat with white nape and cheeks. It is an average 30 cm in length and has a wingspan of 44 cm.

The average length of a **Papuan King Parrot** is 36 cm, including a long broad tail.



- 7 Express the length of a Blue-faced Honeyeater as a fraction of the length of a Papuan King Parrot. Simplify your answer.
- 8 Express the wingspan of a Blue-faced Honeyeater as a percentage of the wingspan of a Papuan Eagle (157 cm). Give your answer correct to 1 decimal place.
- 9 The mass of a Blue-faced Honeyeater is approximately $\frac{1}{20}$ the mass of a Papuan Eagle (2 000 g). What is the mass of a Blue-faced Honeyeater?

The **Forest Kingfisher** has blue wings, head and tail with white breast, abdomen and nape. It has an average length of 23.5 cm.

The male **Golden Monarch Flycatcher** is a bright golden colour with sharply delineated black cheeks and throat, primary wing feathers and tail. It has an average length of 13.5 cm.

- 10 Express the average length of a Golden Monarch Flycatcher as a fraction of the average length of a Forest Kingfisher. Simplify your answer.
- 11 Express the average length of a Forest Kingfisher as a percentage of the average length of a Golden Monarch Flycatcher. Give your answer correct to 1 decimal place.
- 12 Express the length of a King Bird-of-paradise (15 cm) as a percentage of the length of a Forest Kingfisher. Give your answer correct to 1 decimal place.

There are 35 species of bird-of-paradise found in New Guinea. They include the **Superb Bird-of-paradise**, the **Emperor Bird-of-paradise** and the **Yellow Breasted Bird-of-paradise**. The average length of an **Emperor Bird-of-paradise** is 33 cm.

- 13 A Yellow Breasted Bird-of-paradise is approximately $\frac{2}{3}$ the length of an Emperor Bird-of-paradise. How long is a Yellow Breasted Bird-of-paradise?
- 14 A Superb Bird-of-paradise is approximately 80% of the length of an Emperor Bird-of-paradise. How long is a Superb Bird-of-paradise? Give your answer to the nearest centimetre.
- 15 A Raggiana Bird-of-paradise is more than twice the length of a King Bird-of-paradise (15 cm long). The King Bird-of-paradise is approximately 44% of the length of the Raggiana Bird-of-paradise. How long is a Raggiana Bird-of-paradise? Give your answer to the nearest centimetre.

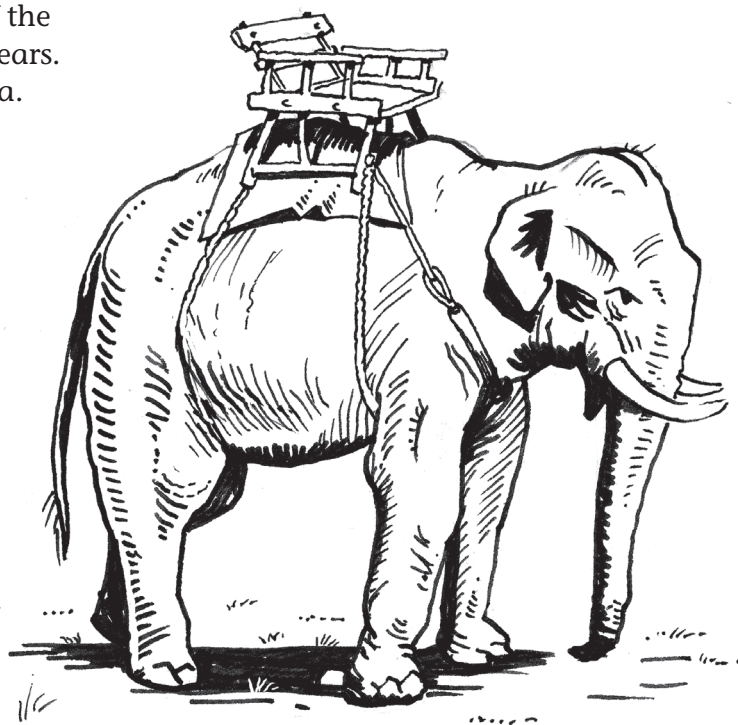


The Asian elephant is a smaller cousin of the African elephant, with relatively smaller ears. It is the largest terrestrial mammal in Asia.

The elephant's massive size and strength made it a reliable beast of burden. It is a national symbol of Thailand and has served many roles in that country's long history. It has been used in timber logging, as a war machine and in processions and ceremonies.

In 2010, it was estimated that only about 2 000 wild elephants remained in the forests of Thailand. It was estimated there were approximately 3 000 domesticated elephants.

Today, domesticated elephants are more likely to be seen carrying tourists or performing in some form of entertainment than working in the jungle or on farms.



- 1 Around 1900 it was estimated that there were at least 100 000 domesticated elephants working in Thailand. By 1952, that number had dropped to around 13 400.

In 2010, it was estimated that there were only 3 000 domesticated elephants and 2 000 wild elephants in Thailand.

Express the number of domesticated elephants in Thailand in 2010 as a percentage of

- a the number of domesticated elephants in 1900.
- b the number of domesticated elephants in 1952.
Answer correct to 1 decimal place.
- c the total number of elephants in 2010.
- d the number of wild elephants in 2010.

- 2 Tan is a mahout (an elephant handler). He is 1.68 m tall and has a mass of 68 kg.

His elephant is 2.8 m tall and has a mass of 4.76 tonnes (4 760 kg).

- a Express Tan's height as a percentage of the height of his elephant.
- b Express Tan's mass as a percentage of the mass of his elephant.
Answer correct to 1 decimal place.

- 3 Tan's elephant eats around 135 kg of plant matter each day.

Express this as a percentage of the mass of his elephant.
Answer correct to 1 decimal place.



Once they are born, baby elephants enjoy a 3 to 5 year childhood before their training begins. They learn to push, carry and stack logs as well as how to walk in a procession. Working elephants have a career of about 50 years.

- 4 Tan began training his elephant when it was 4 years old.
The training took 5 years.
Tan and the elephant then began work.
His elephant is now 25 years old.
- a How many years has the elephant been working?
 - b What percentage of its life did the elephant spend in training?
 - c What percentage of the elephant's life has it been working?
- 5 The law in Thailand requires working elephants to be retired at age 61 after working for 50 years. They are usually released into the wild.
An elephant at a sanctuary is 72 years old.
What percentage of its life did the elephant spend working?
Answer correct to 1 decimal place.

Each year, the town of Surin in north-eastern Thailand hosts an **Elephant Round-up Festival**. More than 300 domesticated elephants are brought to the town by their mahouts. The festival lasts 10 days. The elephants demonstrate their skills and participate in parades and battle re-enactments.

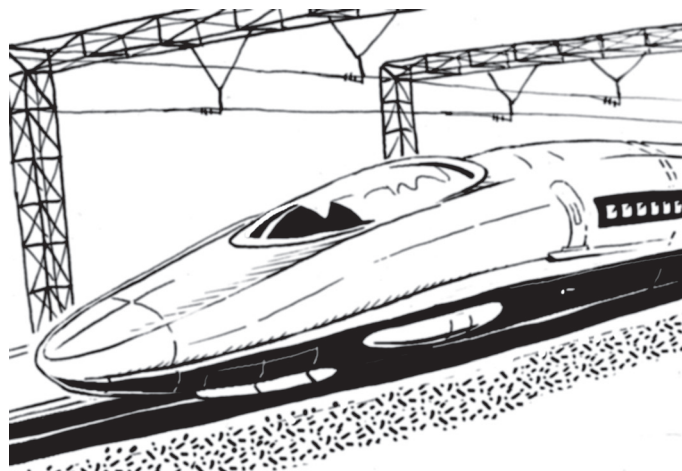
- 6 The average elephant drinks 160 litres of water each day.
- a How many kilolitres of water are needed for each elephant for 10 days?
 - b How many kilolitres of water are needed for 300 elephants for 10 days?
- 7 The average elephant has a mass of 4 800 kg.
It eats 130 kg of plants and vegetables each day.
- a What is the total mass of plants and vegetables the average elephant will eat during the 10 days of the festival?
 - b Express this as a percentage of the mass of the elephant.
Answer correct to 1 decimal place.
- 8 A female elephant stands 2.3 metres in height and has a normal mass of 3.2 tonnes (3 200 kg). She gave birth to a baby. Shortly after birth, the baby stood 90 centimetres high and had a mass of 90 kg.
- a Express the baby's height as a percentage of its mother's height.
Answer correct to the nearest whole number.
 - b Express the baby's mass as a percentage of its mother's mass.
Answer correct to one decimal place.
- 9 A male elephant has a mass of 3.4 tonnes.
Its brain has a mass of 5 kg. Express this as a percentage of the elephant's mass.
Answer correct to two decimal places.

Bullet trains in Japan

4

Japan's high speed trains are called **Shinkansen** (bullet trains). The first bullet train was operated between Tokyo and Osaka in 1964, the year the Olympic Games were held in Tokyo. Today bullet trains run between many cities in Japan. On some lines, trains run every 5 to 10 minutes.

The fastest trains are called 'Nozomi'. They connect major cities like Tokyo, Nagoya, Osaka and Hiroshima, making few stops in between.



- 1 A *Nozomi* train leaves Tokyo at 7:50 am and arrives at Shin-Osaka at 10:29 am. It makes only 4 stops in the 552 kilometre journey.

How long does the journey take?

Answer in hours and minutes.

- 2 A slower *Hikari* train leaves Tokyo at 9:33 am and arrives at Shin-Osaka at 12:30 pm. It makes 6 stops between Tokyo and Shin-Osaka.

a How long does the journey take? Answer in hours and minutes.

b Convert your answer to hours, correct to 2 decimal places.

- 3 A *Nozomi* train leaves Kyoto at 3:45 pm. The 380 kilometre journey to Hiroshima takes 1 hour and 36 minutes. At what time does this train arrive in Hiroshima?

In the following questions you will need the rule:

Average speed = distance travelled ÷ time taken or **Speed = $\frac{\text{Distance}}{\text{Time}}$**

- 4 A train takes 2 hours 18 minutes to travel from Tokyo to Kyoto, a distance of 513 kilometres.

a Convert the time to hours. Give your answer as a decimal.

b Use the rule: $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$ to calculate the average speed of the train, correct to 1 decimal place.

- 5 A train takes 1 hour 25 minutes to travel from Tokyo to Nagano, a distance of 222 kilometres.

a Convert the time to hours, correct to 2 decimal places.

b Calculate the average speed of the train, correct to 1 decimal place.



- 6** A train takes 3 hours 10 minutes to travel from Hiroshima to Nagasaki, a distance of 434 kilometres.

- a** Convert the time to hours, correct to 2 decimal places.
- b** Calculate the average speed of the train, correct to 1 decimal place.



The rule: $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

can be rearranged when you want to calculate the distance travelled or the time taken.

Distance = Speed $\times$ Time and

Time taken = Distance travelled $\div$ Speed

- 7** A train travelled between Tokyo and Karuizawa in 1 hour and 3 minutes. It had an average speed of 139.8 km/h for the journey.
- a** How many hours did the journey take? Give your answer as a decimal.
 - b** What is the distance between Tokyo and Karuizawa, to the nearest kilometre?
- 8** A train left Hiroshima at 9:10 am. It travelled at an average speed of 215.4 km/h and arrived at Tokyo at 1:19 pm.
- a** How many hours did the journey take?
Give your answer correct to 2 decimal places.
 - b** What is the distance between Hiroshima and Tokyo, to the nearest kilometre?
- 9** A train travelled between Tokyo and Shin-Osaka at an average speed of 215 km/h. The two stations are 552 km apart. How long did the journey take?
Give your answer to the nearest minute.
- 10** The distance between Tokyo and Sendai is 351.8 km.
A 'Hayabusa' train completed the journey at an average speed of 219 km/h.
How long did the journey take?
Give your answer to the nearest minute.
- 11** The distance between Shin-Aomori and Tokyo is 713.7 km.
A train departed Shin-Aomori at 10:35 am and travelled to Tokyo at an average speed of 203.9 km/h.
- a** How long did the journey take?
Give your answer to the nearest minute.
 - b** At what time did this train arrive at Tokyo?



Justine and Melissa were visiting Vietnam. They saw many street markets where fresh fruit and vegetables were sold. The prices were all marked in the local currency. The local money is known as Vietnamese dong (VND). The prices looked very high. However, the exchange rate was one Australian dollar was approximately equal to 21 000 VND.



- 1 Justine decided to buy some fruit. The prices were based on one kilogram. Justine purchased:

1 kg of bananas @ 15 000 VND per kg
 800 grams (0.8 kg) of oranges @ 24 000 VND per kg
 500 grams (0.5 kg) of rambutans @ 25 000 VND per kg

- a Calculate the cost of each item.
- b Calculate the total cost of Justine's purchase.
- c Justine paid using a 50 000 VND note. How much change should she receive?

- 2 The next day Melissa purchased:

900 grams of mangoes @ 30 000 VND per kg
 700 grams of lychees @ 50 000 VND per kg
 800 grams of apples @ 16 000 VND per kg

- a Calculate the cost of each item.
- b Calculate the total cost of Melissa's purchase.
- c Melissa gave the vendor 80 000 VND. How much change should she receive?

- 3 Their friend Mrs Linh was doing her shopping. She purchased:

600 grams of sweet potatoes @ 16 000 VND per kg
 500 grams of beans @ 35 000 VND per kg
 300 grams of eggplant @ 17 000 VND per kg
 200 grams of morning glory @ 25 000 VND per kg

Calculate the total cost of Mrs Linh's shopping.



- 4** The currency exchange rate was one Australian dollar was approximately equal to 21 000 VND. To convert from Vietnamese dong to Australian dollars, you divide by 21 000.

Use this rate to convert these amounts to Australian dollars.
Answer to the nearest cent.

- a** 25 000 VND
- b** 40 000 VND
- c** 60 000 VND
- d** 75 000 VND
- e** 100 000 VND
- f** 130 000 VND
- g** 170 000 VND
- h** 250 000 VND

- 5** Convert the total cost of Justine's purchase (Question 1) to Australian dollars.
- 6** Convert the total cost of Melissa's purchase (Question 2) to Australian dollars.
- 7** Convert the total cost of Mrs Linh's purchase (Question 3) to Australian dollars.
- 8** Thang was buying fruit for his family. He purchased:

800 grams of bananas @ 16 000 VND per kg
 1.2 kg of red papaya @ 14 000 VND per kg
 1.4 kg of dragon fruit @ 26 000 VND per kg
 600 grams of rambutans @ 25 000 VND per kg

- a** Calculate the total cost of Thang's shopping in VND.
 - b** Convert your answer to Australian dollars. Answer to the nearest cent.
- 9** Justine saw a woman selling flowers from a bicycle. She decided to purchase some flowers to give to Mrs Linh. The flowers cost 65 000 VND. Convert this amount to Australian dollars.





Matthew was reading about some of the tall buildings in Asia.

The **Jin Mao Tower** in Shanghai (China) is 420 metres high.

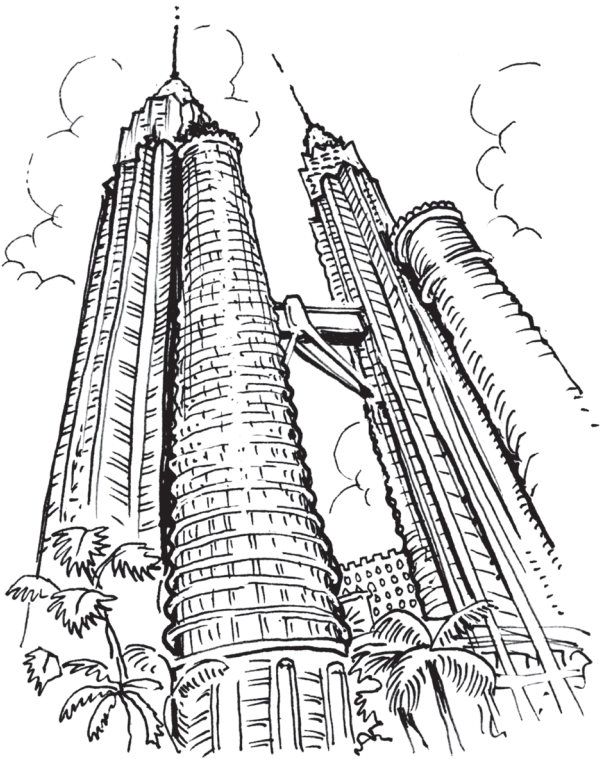
The **Petronas Twin Towers** in Kuala Lumpur (Malaysia) is 452 metres high.

Taipei 101 in the capital of Taiwan is 509 metres high.

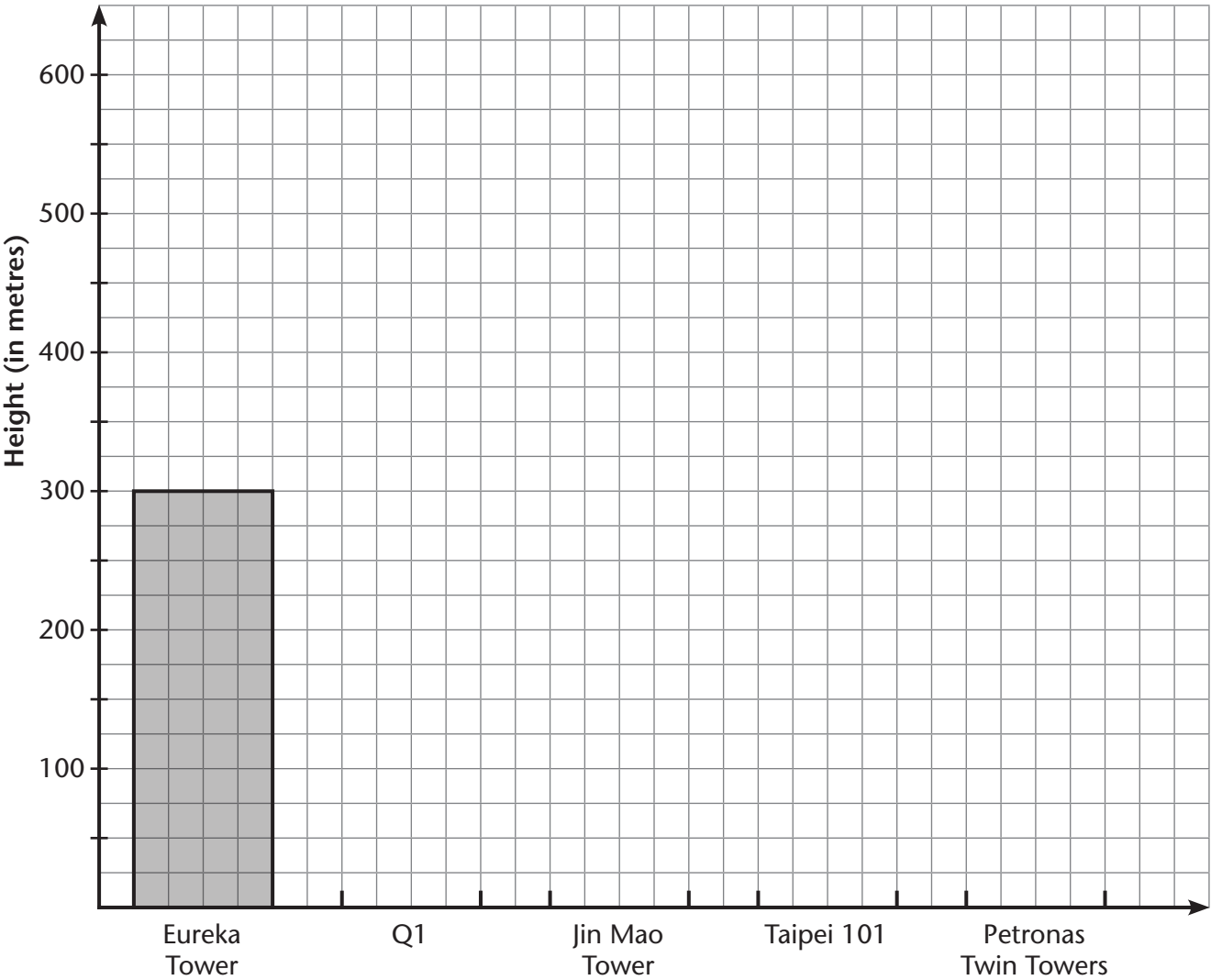
Matthew wanted to compare the heights of these buildings with the heights of the **Eureka Tower** in Melbourne (297 metres high) and **Q1** on the Queensland Gold Coast (323 metres high).

First, he constructed a column graph of the heights.

- 1 Complete this column graph representing the heights of these buildings.



Heights of tall buildings





Matthew wanted to compare the ratios of the heights.

- 2 To simplify the calculations, round each height to the nearest 10 metres.

Building	Height	Height to the nearest 10 metres
Jin Mao Tower	420 m	420 m
Petronas Twin Towers	452 m	
Taipei 101	509 m	
Eureka Tower	297 m	
Q1	323 m	

Use your answers in question 2 as the heights to answer the following questions. Simplify your answers where possible.

To simplify a ratio, divide both terms by any common factors.

A common factor will divide both numbers without a remainder.

- 3 What is the ratio of the heights of these pairs of buildings?

- a Eureka Tower to Petronas Twin Towers
- b Eureka Tower to Taipei 101
- c Q1 to Jin Mao Tower
- d Q1 to Petronas Twin Towers
- e Q1 to Taipei 101
- f Jin Mao Tower to Eureka Tower
- g Petronas Twin Towers to Eureka Tower
- h Taipei 101 to Jin Mao Tower

- 4 Express the height of the Eureka Tower as a fraction of the height of the Petronas Twin Towers.

- 5 Express the height of the Petronas Twin Towers as a fraction of the height of the Eureka Tower.

- 6 Express the height of the Eureka Tower as a fraction of the height of Q1.

- 7 The mass of the Eureka Tower is 200 000 tonnes. Taipei 101 has a total mass of 700 000 tonnes.

- a What is the ratio of the mass of the Eureka Tower to the mass of Taipei 101?
- b What is the ratio of the mass of Taipei 101 to the mass of the Eureka Tower?
- c Express the mass of Taipei 101 as a fraction of the mass of the Eureka Tower.





The table shows the population (in thousands) for a number of countries. The population is shown at intervals of five years from 1990 to 2010.

Table 1: Population (in thousands)

Country	1990	1995	2000	2005	2010
Australia	17 096	18 118	19 164	20 404	22 268
India	837 785	964 486	1 053 898	1 140 043	1 224 614
Indonesia	184 346	199 400	213 395	227 303	239 871
Philippines	61 629	69 255	77 310	85 546	93 261
Thailand	57 072	59 650	63 155	66 698	69 122

(United Nations ESCAP 2011)

Answer these questions using the data in this table.

- 1
- What was the population of Australia in 1990?
- 2
- What was the population of India in 2000?
- 3
- List the countries in increasing order of population size (smallest to largest) in 2010.
- 4
- a

What was the increase in the population of Australia from 1990 to 1995?

b

Express this increase as a percentage of the population in 1990.
Give your answer correct to 2 decimal places.

c

What was the average percentage increase per year from 1990 to 1995?
Give your answer correct to 1 decimal place.
- 5
- a

Calculate the average percentage increase in population per year from 1990 to 1995 for each country. Round each answer correct to 1 decimal place.
Record your answers in the first column of the table below.

Table 2: Average population increase per year

Country	1990–1995	1995–2000	2000–2005	2005–2010
Australia		1.2%	1.3%	
India			1.6%	1.5%
Indonesia		1.4%		1.1%
Philippines		2.3%		1.8%
Thailand		1.2%	1.1%	

- b
- Complete the remaining columns in the table.



Population growth in the region has been steadily declining over the last two decades. In 2010 the Asia-Pacific region’s annual population growth rate averaged 1.0%.

- 6 Which countries listed in the table had an annual population growth rate of more than 1.0% in the period 2005–2010?
- 7 Which countries had an annual population growth rate of less than 1.0% in this period?

The following table shows the birth rates for these countries, expressed as the number of births per 1 000 of the population.

Table 3: Birth rate (births per 1 000 population)

Country	1990–1995	1995–2000	2000–2005	2005–2010	2010
Australia	14.7	13.4	12.7	13.6	13.7
India	30.0	27.2	24.8	23.1	22.2
Indonesia	24.5	21.9	21.0	19.1	18.2
Philippines	31.8	30.2	29.0	25.9	25.1
Thailand	18.0	15.6	14.1	12.9	12.1

(United Nations ESCAP 2011)

- 8 In the period 1990 to 1995, which country had the highest birth rate?
- 9 In the period 1990 to 1995, which country had the lowest birth rate?
- 10 In 2010, which country had the highest birth rate?
- 11 In 2010, which country had the lowest birth rate?
- 12 Which country did not have a consistently declining birth rate from 1990 to 2010?
- 13 Calculate the number of births in the following countries in 2010.

Use the birth rates shown in Table 3 and the corresponding populations shown in Table 1. Round your answers to the nearest hundred.

- a Australia
- b India
- c Indonesia





The table shows the average number of males for every 100 females in the population of a number of countries. The population is shown at intervals of five years from 1990 to 2010.

Population ratio (males per 100 females)

Country	1990	1995	2000	2005	2010
Australia	99	99	98	99	99
Cambodia	91	93	94	95	96
China	107	107	107	108	108
India	108	108	108	107	107
Indonesia	100	100	100	100	99
Malaysia	103	103	103	103	103
Vietnam	95	96	97	97	98



(United Nations ESCAP 2011)

- 1 In 2010, were there more males or females in the population of Australia?
- 2 In 1990, which country had the highest number of males per 100 females?
- 3 In 1990, which country had the lowest number of males per 100 females?
- 4 In 2010, which country had the highest number of males per 100 females?
- 5 In 2010, which country had the lowest number of males per 100 females?
- 6 What was the ratio of males to females in Australia in 1990?
- 7 a What was the ratio of males to females in Indonesia in 1990?
b Simplify this ratio.
- 8 a What was the ratio of males to females in Vietnam in 1995?
b Simplify this ratio.
- 9 What was the ratio of males to females in Cambodia in 2005?
Give your answer in its simplest form.
- 10 What was the ratio of males to females in China in 2010?
Give your answer in its simplest form.
- 11 Which country had no change in the number of males per 100 females in each survey?
- 12 Which country had an increase in the number of males per 100 females in each survey?



- 13 In 1990, the population of Indonesia was approximately 184 346 000. How many of these people were male? Give your answer to the nearest thousand.
- 14 In 2010, the population of Australia was approximately 22 268 000. How many of these people were female? Give your answer to the nearest thousand.

The following table shows the average number of male children for every 100 female children in the population of a number of countries. The population is shown at intervals of five years from 1990 to 2010.

Children (ratio of males per 100 females)

Country	1990	1995	2000	2005	2010
Australia	105	105	105	105	105
Cambodia	102	103	103	104	104
China	108	111	115	119	121
India	108	108	109	109	109
Indonesia	103	103	103	104	104
Malaysia	105	105	105	106	106
Vietnam	103	103	104	104	105



(United Nations ESCAP 2011)

- 15 In 1990, which country had the lowest number of male children per 100 female children?
- 16 In 1990, which two countries had the highest number of male children per 100 female children?
- 17 In 2010, which country had the highest number of male children per 100 female children?
- 18 In 2010, which two countries had the lowest number of male children per 100 female children?
- 19 What was the ratio of male children to female children in Australia in 1995? Simplify your answer.
- 20 What was the ratio of male children to female children in China in 2000? Give your answer in its simplest form.
- 21 What was the ratio of male children to female children in Indonesia in 2010? Give your answer in its simplest form.
- 22 Which country had no change in the number of male children per 100 females in each survey?
- 23 Which country had an increase in the number of male children per 100 females in each survey?



The Islamic architecture of the Moghul period in India is a fusion of Arab, Central Asian, Persian and Hindu elements. The best known example is the Taj Mahal which was completed in 1648.

- 1 Kathryn and Adam were visiting the Taj Mahal. They were amazed at the symmetry and beauty of this building. Adam noticed this tessellation in a section of paving. The pattern is formed by repeatedly translating a unit design on a square.

- a One square is outlined on the diagram. Use a coloured pen to outline the grid squares on the tessellation.

One unit square of the pattern is shown in the second diagram.

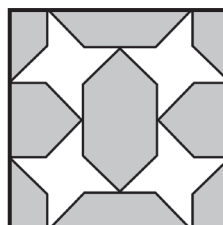
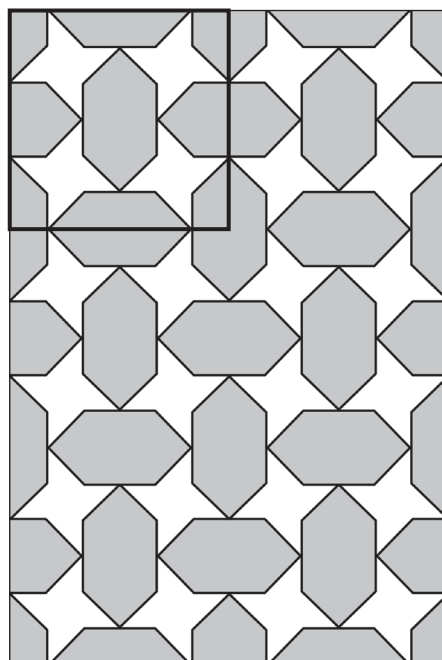
- b How many **axes of symmetry** are there in the unit square?

An axis of symmetry is a line where the figure can be folded or reflected so that the two halves match exactly.

Draw the axes of symmetry on the diagram.

- c What is the order of **rotational** symmetry of the unit square?

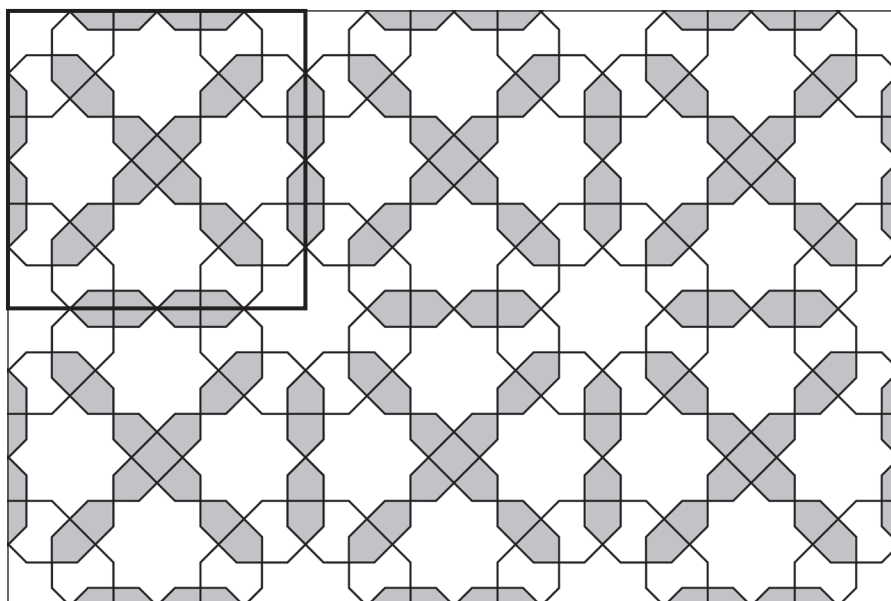
A figure has rotational symmetry if it exactly matches its original position more than once when it is turned through one revolution.



- 2 At another mausoleum near Agra they found a tiling pattern based on this tessellation. It is also formed by repeatedly translating a square unit.

- a One square is outlined on the diagram.

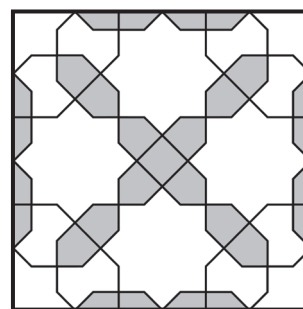
Use a coloured pen to outline the grid squares on the tessellation.



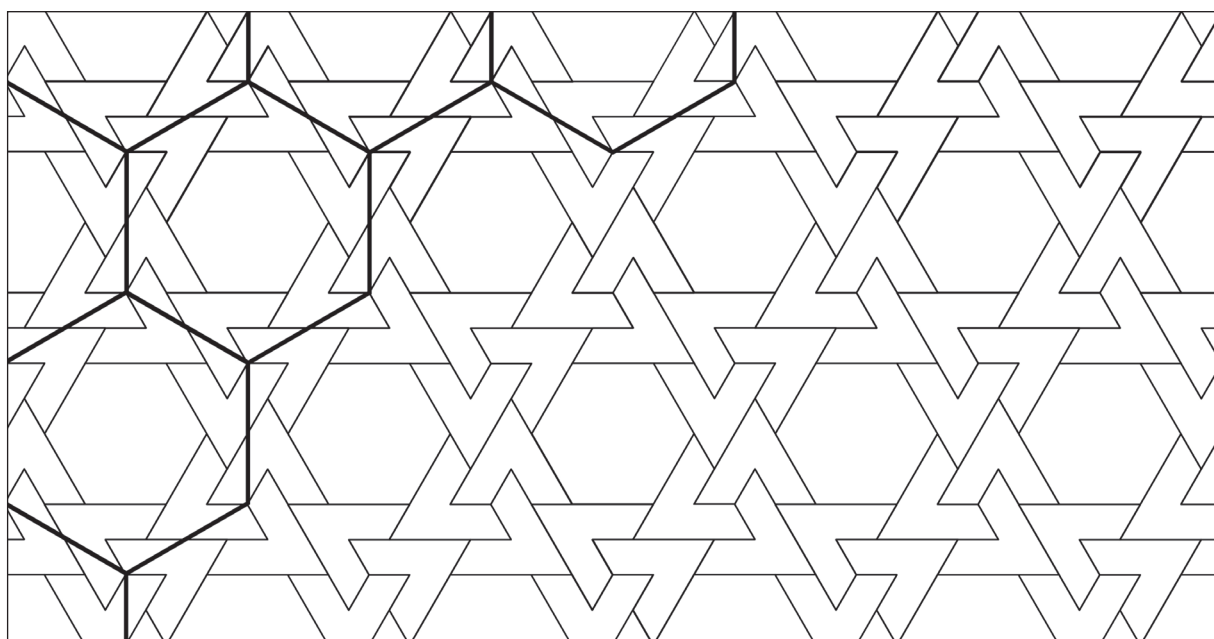


One unit square of the pattern is shown in this diagram.

- b** How many **axes of symmetry** are there?
Draw the axes of symmetry on the diagram.
- c** What is the order of **rotational symmetry**?



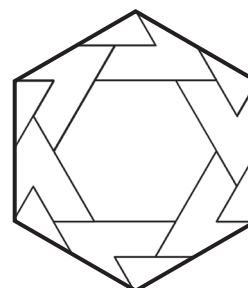
- 3** Kathryn and Adam later visited the tomb of the Moghul Emperor Akbar. Here they saw a tessellation based on a hexagon. Part of the hexagon grid is outlined on the diagram.



- a** How many axes of symmetry does a regular hexagon (with no design) have?
- b** What is the order of rotational symmetry of a regular hexagon?

One unit hexagon of the pattern is shown in this diagram.

- c** How many **axes of symmetry** are there?
- d** What is the order of **rotational symmetry**?

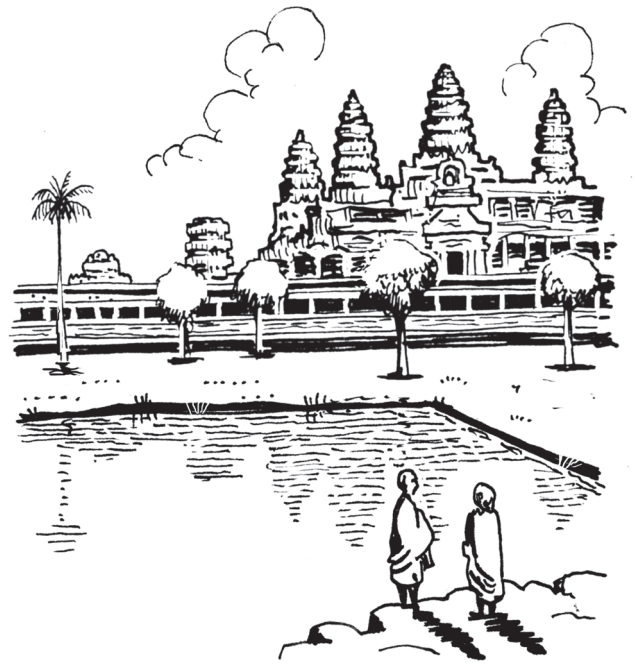


The city of Angkor was the capital of the Khmer Empire. Its ruins are located near the modern city of Siem Reap in Cambodia.

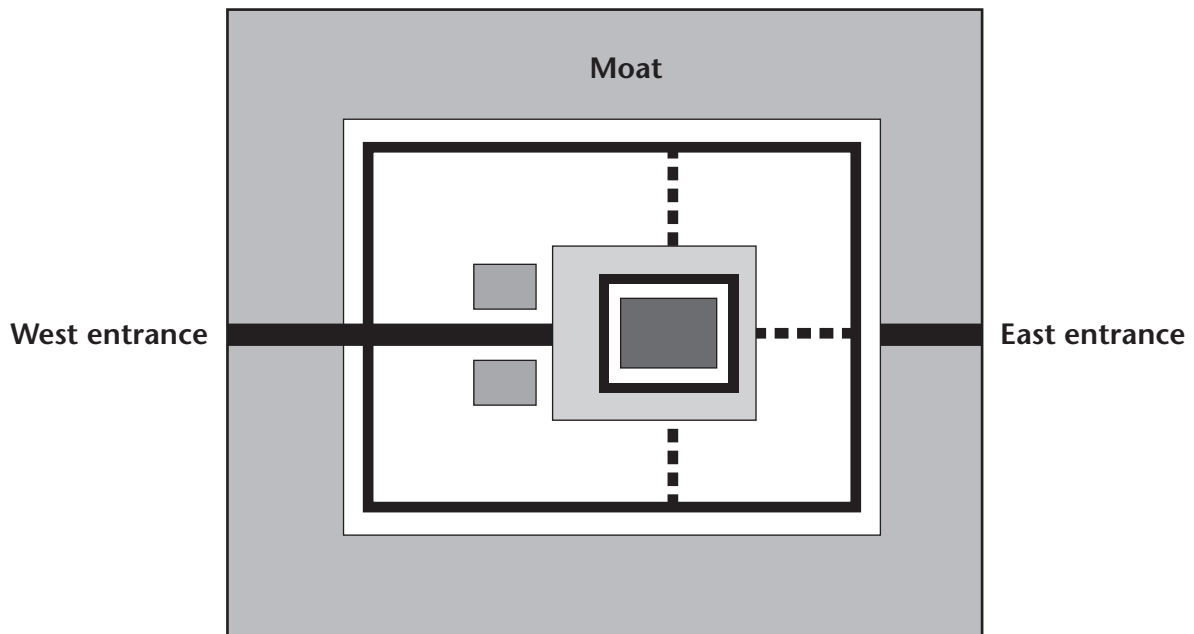
The temples of Angkor were built during the period 802 to 1432. During this time, the Khmer Empire was one of the great powers of South East Asia. At the height of its power, the Khmer Empire ruled over a vast territory covering large parts of modern-day Cambodia, Laos, Thailand, Myanmar (Burma) and southern Vietnam.

Angkor Wat (meaning *City Temple*) is the best known of the temples in this area. It was built as a Hindu temple during the reign of Suryavarman II (1112 to 1152). It was later restored for use as a Buddhist shrine in the 16th century.

Angkor Wat is surrounded by a moat, 190 metres wide, that forms a giant rectangle measuring 1.5 km by 1.3 km. The rectangular wall around the enclosure measures 1025 m by 800 m, and has a gate in each side.



Layout of Angkor Wat



(Not drawn to scale)

- 1 Calculate the perimeter of the moat.
- 2 Calculate the perimeter of the walled enclosure inside the moat.
- 3 What is the area of the enclosure?
- 4 **a** Calculate the area of the large rectangle, including the moat, in square kilometres.
b Convert your answer to square metres.
- 5 What are the dimensions of the area inside the moat?
- 6 Calculate the area of the moat.



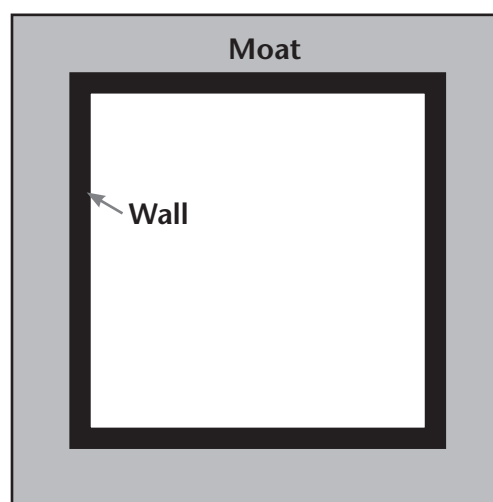
The main entrance to **Angkor Wat** is on the western side. An avenue 475 m long and 9.5 m wide leads from the main entrance to the central temple. The central temple complex consists of three storeys. The corners of the second and third storeys are marked by towers. The central tower rises 31 m above the third storey and 55 m above the ground.

- 7 What is the area of the avenue from the main entrance to the central temple?
- 8 How high above the ground is the third storey?

There are more than 100 temples in the area around Angkor. Many of the temples were left to the jungle for centuries and today are found in various stages of decay.

- 9 **Banteay Kdai** is a massive temple built in the 12th century. It is surrounded by four concentric walls. The outer wall measures 700 metres $\times$ 500 metres.
 - a What is the perimeter of the outer wall?
 - b Calculate the area enclosed by the outer wall.
- 10 Near Banteay Kdai is a large pool, the Pool of Ablutions (**Sras Srang**). This rectangular pool is 400 metres wide. It covers an area of 320 000 square metres.
How long is the pool?
- 11 Another temple is **Preah Khan** (*Sacred Sword*). The temple is surrounded by a wall. The area inside the wall is 56 hectares. One side of the rectangle is 800 metres long.
 - a Convert 56 hectares to square metres. Hint: 1 hectare = 10 000 square metres.
 - b How long is the other side of the wall?

- 12 The fortified city of **Angkor Thom** was enclosed by a square wall 8 metres high and 12 kilometres in length.
This was encircled by a moat 100 metres wide.
 - a What was the length of the wall on one side of the city?
 - b What was the length of one side of the moat?
 - c What was the area of the city inside the wall?
 - d What was the area of the city and the moat?
 - e What was the area of the moat?



(Not drawn to scale)



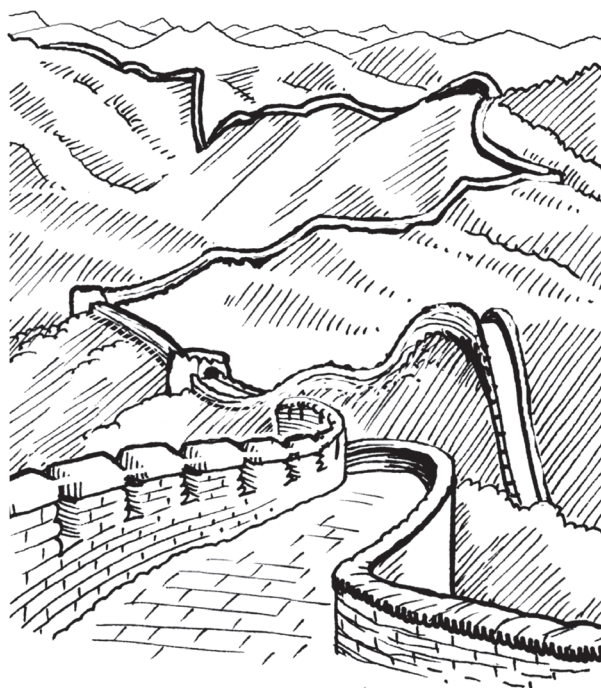
The Great Wall in northern China is one of the great human-made wonders of the world. Like a gigantic dragon, the Great Wall winds up and down across deserts, grasslands and mountains.

The first sections were constructed more than 2 000 years ago by independent kingdoms to keep out marauding nomads. These sections were later rebuilt and joined together.

The wall as it is known today dates from the Ming Dynasty (1368–1644) when a determined effort was made to revitalise the whole project. This time, the wall was faced with some 60 million cubic metres of bricks and stone slabs.

The Great Wall stretched approximately 8 850 kilometres from east to west.

Unfortunately, some of the sections are now in ruins or have disappeared.



A comprehensive archaeological survey has concluded that the wall of the Ming era measured 8 850 km. This is made up of 6 259 km of actual wall, 359 km of trenches and 2 232 km of natural defensive barriers such as hills and rivers.

- 1 What percentage, correct to 1 decimal place, of the Ming-era wall was
 - a actual wall?
 - b trenches?
 - c natural defensive barriers?

- 2 The average height of the Ming-era Great Wall measured 10.1 metres and the average width was about 4.6 metres. Assume the wall was a rectangular prism. Use these dimensions to calculate

- a the area of a one face of the wall 1 000 metres long.
- b the volume of a 1 kilometre section of the wall.
Answer in cubic metres.

- 3 In the Ming era, the wall was faced with approximately 60 million cubic metres of bricks and stone slabs. The length of actual wall was 6 259 kilometres.

What was the average volume of bricks and stone slabs used for each kilometre of the wall?
Answer to the nearest cubic metre.

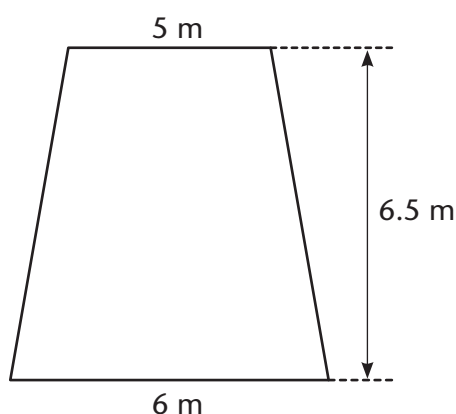


One of the best-preserved sections of the Great Wall is situated at Badaling, about 70 kilometres north of Beijing. Most tourists visit this section of the wall.

Made of stone and bricks from the hills, this portion of the Great Wall is 7.8 metres high and 5 metres wide. The inside has been formed by tamping earth and small stones, which makes the wall very firm and strong.

4 Use these dimensions to calculate

- a** the area of a one face of the wall 500 metres long.
- b** the volume of a 1 kilometre section of the wall (assuming it is a rectangular prism).
Give your answer in cubic metres.



One of the most striking sections of the Ming Great Wall is where it climbs extremely steep slopes. In one section, the cross-section is a trapezium. The wall has an average height of 6.5 metres. It is 6 metres across the bottom, narrowing to 5 metres across the top.

- 5**
 - a** Calculate the area of a trapezium 6.5 m high with parallel sides measuring 5 m and 6 m.
 - b** Calculate the volume of a trapezoidal prism 10 m long, 6.5 m high with parallel sides measuring 5 m and 6 m.
 - c** Calculate the volume of a 5 kilometre section of this part of the Great Wall.
Give your answer in cubic metres.

Over time, the Great Wall has slowly decayed due to the forces of nature. People have removed some of the stone to use in other structures. Today, less than 30% of the Great Wall still stands in good condition.

- 6** What length is 30% of 8 850 km?
Give your answer correct to the nearest kilometre.





Australia has three time zones (ignoring daylight saving in summer):

- **Eastern Standard Time** is used in Queensland, New South Wales (excluding Broken Hill), Victoria, Tasmania and the ACT.
- **Central Standard Time** is used in South Australia, the Northern Territory and Broken Hill.
- **Western Standard Time** is used in Western Australia.

Eastern Standard Time is 30 minutes ahead of **Central Standard Time**.

It is 30 minutes earlier in Adelaide than Brisbane.

Eastern Standard Time is 2 hours ahead of **Western Standard Time**.

It is 2 hours earlier in Perth than Brisbane.

Papua New Guinea is also in this time zone.

- 1 When it is 8:00 am in Sydney, what is the time in
 - a Brisbane?
 - b Adelaide?
 - c Perth?
 - d Port Moresby (Papua New Guinea)?
- 2 When it is 4:30 pm in Mount Hagan (Papua New Guinea), what is the time in
 - a Sydney?
 - b Darwin?

Central Standard Time is 30 minutes behind **Eastern Standard Time**.

It is 30 minutes later in Melbourne than Adelaide.

Central Standard Time is 1 hour and 30 minutes ahead of **Western Standard Time**.

It is 1 hour and 30 minutes earlier in Perth than Adelaide.

Japan, Korea and the eastern part of **Indonesia** are also in this time zone.

- 3 When it is 12 noon in Adelaide, what is the time in
 - a Melbourne?
 - b Perth?
 - c Tokyo (Japan)?
 - d Seoul (South Korea)?
- 4 When it is 1:30 pm in Osaka (Japan), what is the time in
 - a Adelaide?
 - b Hobart?

Western Standard Time is 2 hours behind **Eastern Standard Time**.

It is 2 hours later in Sydney than Perth.

Western Standard Time is 1 hour and 30 minutes behind **Central Standard Time**.

It is 1 hour and 30 minutes later in Adelaide than Perth.

China, Malaysia and the **Philippines** are also in this time zone.

- 5 When it is 3:00 pm in Perth, what is the time in
 - a Brisbane?
 - b Adelaide?
 - c Beijing (China)?
 - d Manilla (Philippines)?



- 6 When it is 11:30 am in Kuala Lumpur (Malaysia), what is the time in
a Perth? **b** Sydney?
- 7 Julie was holidaying in Japan.
 At 8:30 am she phoned her family in Adelaide.
 What was the time in Adelaide when Julie phoned?
- 8 Lei called his family in China when
 the time was 10:00 pm in Sydney.
 What was the time in China when he called?
- 9 At what time should I make an
 international telephone call from
 Melbourne in order to call
a Japan at 10:00 am?
b China at 6:00 pm?



The following questions use **24-hour time**.

24-hour times are used on many airline schedules.

24-hour time starts from midnight (00:00 hours).

All times are written using 4 digits.

The first two digits indicate the number of hours after midnight.

The last two digits indicate the number of minutes.

For example, 14:30 is equal to 2:30 pm.

- 10 When it is 14:30 in Melbourne, what is the time in
a Brisbane? **b** Port Moresby (Papua New Guinea)?
- 11 When it is 11:15 in Perth, what is the time in
a Adelaide? **b** Beijing (China)?
- 12 When it is 21:40 in Adelaide, what is the time in
a Sydney? **b** Tokyo (Japan)?
- 13 A plane leaves Brisbane at 14:20.
 It takes 3 hours 5 minutes to fly from Brisbane to Port Moresby (Papua New Guinea).
 What is the local time when this plane arrives in Port Moresby?
- 14 A plane leaves Perth at 07:50.
 It takes 5 hours 40 minutes to fly from Perth to Kuala Lumpur (Malaysia).
 What is the local time when this plane arrives in Kuala Lumpur?
- 15 A plane leaves Beijing (China) at 08:40.
 It takes 13 hours to fly from Beijing to Sydney.
 What is the local time when this plane arrives in Sydney?
- 16 Natalie caught a plane from Kuala Lumpur (Malaysia) to Adelaide.
 The flight took 7 hours and 20 minutes, non-stop.
 The plane left Kuala Lumpur at 23:55.
 At what time did the plane arrive in Adelaide?



Australia has three time zones (ignoring daylight saving in summer):

- **Eastern Standard Time** is used in Queensland, New South Wales (excluding Broken Hill), Victoria, Tasmania and the ACT.
- **Central Standard Time** is used in South Australia, the Northern Territory and Broken Hill.
- **Western Standard Time** is used in Western Australia.

The time in **New Zealand** is 2 hours ahead of Australian Eastern Standard Time.
It is 2.5 hours ahead of South Australia and 4 hours ahead of Western Australia.

- 1 When it is 9:30 am in Sydney, what is the time in Auckland (New Zealand)?
- 2 When it is 11:00 am in Christchurch (New Zealand), what is the time in
 - a Brisbane?
 - b Melbourne?
 - c Perth?
 - d Adelaide?

The time in **Thailand, Vietnam and Cambodia** is 3 hours behind Australian Eastern Standard Time.

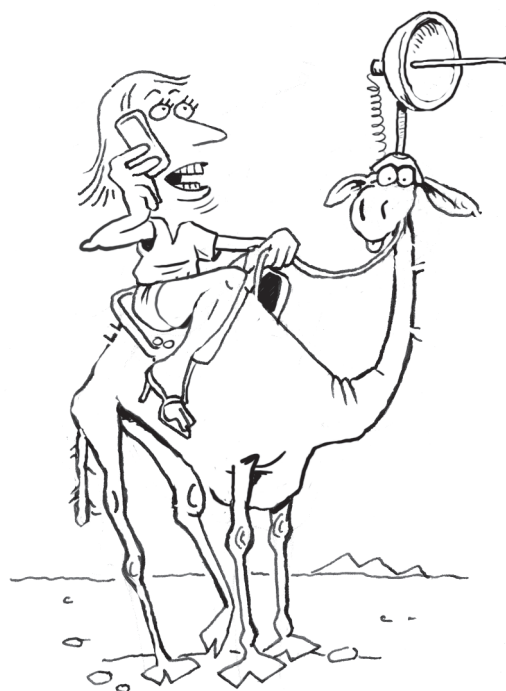
It is 2.5 hours behind South Australia and 1 hour behind Western Australia.

- 3 When it is 10:00 am in Melbourne, what is the time in Bangkok (Thailand)?
- 4 When it is 3:00 pm in Hanoi (Vietnam), what is the time in
 - a Sydney?
 - b Brisbane?
 - c Adelaide?
 - d Perth?
- 5 Thang lives in Brisbane.
He telephoned his family in Vietnam when it was 2:00 pm in Brisbane.
What was the time in Vietnam when he called?

The time in **India and Sri Lanka** is 2.5 hours behind the time in Western Australia.

The time in India and Sri Lanka is 4 hours behind South Australia and 4.5 hours behind Australian Eastern Standard Time.

- 6 When it is 4:00 pm in Perth, what is the time in Mumbai (India)?
- 7 When it is 11:30 am in Colombo (Sri Lanka), what is the time in
 - a Perth?
 - b Brisbane?
 - c Melbourne?
 - d Adelaide?





- 8** Alison and Graham are watching television at home in Perth. They are watching a live cricket match which is being played in Colombo (Sri Lanka). The time is 3:30 pm in Perth. What is the time in Colombo?

The time in **Bangladesh** is 3.5 hours behind Australian Central Standard Time.

- 9** When it is 4:00 pm in Adelaide, what is the time in Dacca (Bangladesh)?
- 10** When it is 11:30 pm in Dacca (Bangladesh), what is the time in
- a** Adelaide?
 - b** Brisbane?
 - c** Melbourne?
 - d** Perth?
- 11** The Australian Open tennis tournament is being telecast live from Melbourne. It is 3:00 pm in Melbourne. What is the time in
- a** Wellington (New Zealand)?
 - b** Bangkok (Thailand)?
 - c** Delhi (India)?
- 12** Jane was holidaying in Cambodia. At 8:00 pm she phoned her family in Adelaide. What was the time in Adelaide when Jane phoned?
- 13** At what time should I make an international telephone call from Brisbane in order to call
- a** Vietnam at 10:00 am?
 - b** Sri Lanka at 6:00 pm?

The following questions use **24-hour time**.

24-hour time starts from midnight (00:00 hours).

All times are written using 4 digits.

The first two digits indicate the number of hours after midnight.

The last two digits indicate the number of minutes.

For example, 16:20 is equal to 4:20 pm.

- 14** A plane leaves Auckland (New Zealand) at 16:20. It takes 3 hours 30 minutes to fly from Auckland to Sydney. What is the local time when this plane arrives in Sydney?
- 15** A plane leaves Bangkok (Thailand) at 07:45. It takes 8 hours 50 minutes to fly from Bangkok to Melbourne. What is the local time when this plane arrives in Melbourne?
- 16** Casey is flying from Brisbane to Auckland (New Zealand). The flight left Brisbane at 08:20 and is due to arrive in Auckland at 14:35. How long is Casey's flight?



South East Asia’s forests covered 214 million hectares in 2010. They covered 49% of the land area in the sub-region (FAO 2011).

The loss of forest area in South East Asia threatens timber production and biodiversity. Between 1990 and 2010, these forests contracted in size by almost 33 million hectares, an area greater than that of Vietnam.



Table 1: Area of forest and rates of change of forest area in South East Asia

	Forest area 2010 (thousand hectares)	Forest cover (% of land area)	Annual change in forest area (percentage per annum)		
			1990– 2000	2000– 2005	2005– 2010
Cambodia	10 094	57	-1.1	-1.5	-1.2
Indonesia	94 432	52	-1.7	-0.2	-0.7
Lao PDR	15 751	68	-0.5	-0.5	-0.5
Malaysia	20 456	62	-0.4	-0.7	-0.4
Myanmar	31 773	48	-1.2	-0.9	-0.9
Philippines	7 665	26	0.8	0.8	0.7
Thailand	18 972	37	-0.3	-0.1	0.1
Vietnam	13 797	42	2.3	2.2	1.1
SE Asia (total)	214 064	49	-1.0	-0.3	-0.5

(FAO 2011)

- 1 How many hectares of land in Cambodia were forest in 2010?
- 2 How many hectares of land in the Philippines were forest in 2010?
- 3 What percentage of the land in Indonesia was forest in 2010?
- 4 Which countries had an increase in the area of forest cover between 1990 and 2000?
(They had a **positive** annual percentage change.)
- 5 Which countries had an increase in the area of forest cover between 2005 and 2010?
- 6 Which country had the greatest decrease in the area of forest cover
in the period between 1990 and 2000?
- 7 Which country had the greatest decrease in the area of forest cover
in the period between 2005 and 2010?



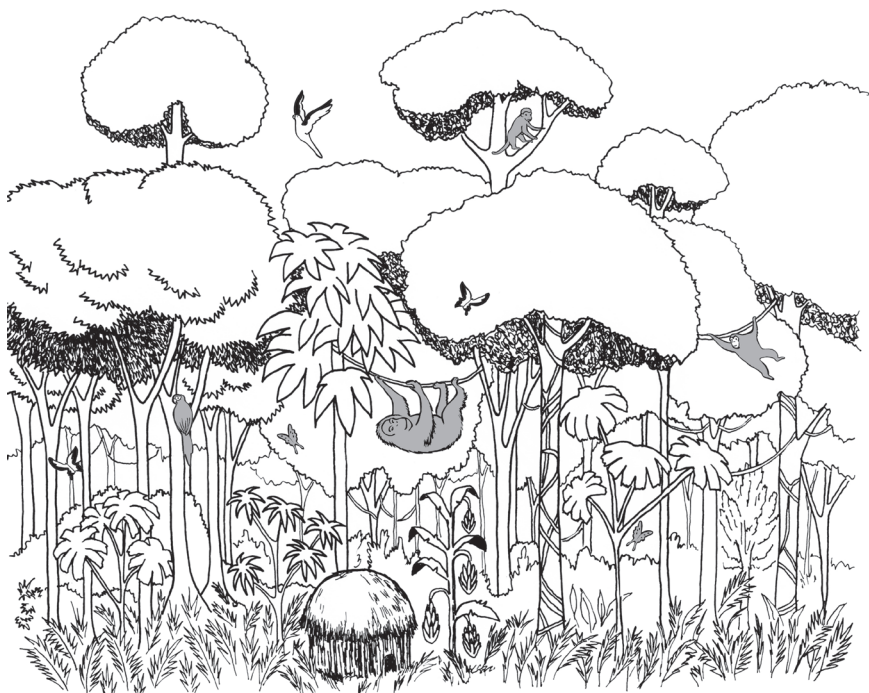
As natural forests have been logged for timber or cleared for farming, efforts have been made to plant new forests. In most countries, the proportion of planted forest remained low in 2010. However, in Thailand and Vietnam planted forests make up a significant proportion of the total forest area.

Table 2: Natural and planted forest area in South East Asia in 2010

	Total forest area (thousand hectares)	Primary forest (thousand hectares)	% Primary forest	Other naturally regenerated (thousand hectares)	Planted forest (thousand hectares)	% Planted forest
Cambodia	10 094	322	3.2	9 703	69	0.7
Indonesia	94 432	47 236	50.0	43 647	3 549	3.8
Lao PDR	15 751	1 490	9.5	14 037	224	1.4
Malaysia	20 456	3 820	18.7	14 829	1 807	8.8
Myanmar	31 773	3 192	10.0	27 593	988	3.1
Philippines	7 665	861	11.2	6 452	352	4.6
Thailand	18 972	6 726	35.5	8 261	3 986	21.0
Vietnam	13 797	80	0.6	10 205	3 512	25.5
SE Asia (total)	214 064	63 992	29.9	135 540	14 533	6.8

(FAO 2011)

- 8 In 2010, how many hectares of land in Cambodia were
- a primary forest?
- b planted forest?
- 9 How many hectares of land in Thailand were
- a primary forest?
- b planted forest?
- 10 What percentage of the total forest area in Thailand was
- a primary forest?
- b planted forest?
- 11 What percentage of the total forest area in Vietnam was
- a primary forest?
- b planted forest?





Alicja and David are planning a holiday in South East Asia. They plan to visit Cambodia and Thailand. In Thailand they want to spend a few days in Bangkok as well as enjoy time on the beach at Koh Samui.

They found the following information in travel brochures.

Bangkok (Thailand)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Av. max. temp. (°C)	32	32	34	35	34	33	33	33	33	32	32	31
Av. min. Temp. (°C)	21	22	25	26	26	26	26	25	25	25	23	21
Av. rain days	1	1	3	6	16	16	17	20	21	17	6	1

Koh Samui (Thailand)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Av. max. temp. (°C)	28	30	31	32	33	32	32	32	32	31	30	29
Av. min. Temp. (°C)	24	25	26	26	26	26	25	25	25	24	24	24
Av. rain days	10	6	5	8	15	14	14	15	16	20	20	14

Phnom Penh (Cambodia)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Av. max. temp. (°C)	32	33	35	35	34	34	33	33	32	31	30	30
Av. min. Temp. (°C)	22	23	24	25	25	25	25	25	24	24	23	22
Av. rain days	3	2	5	9	16	17	20	21	20	24	12	5

- 1 What is the average maximum temperature in Bangkok during January?
- 2 What is the highest average maximum temperature in

a Bangkok?

b Koh Samui?

c Phnom Penh?
- 3 What is the lowest average maximum temperature in

a Bangkok?

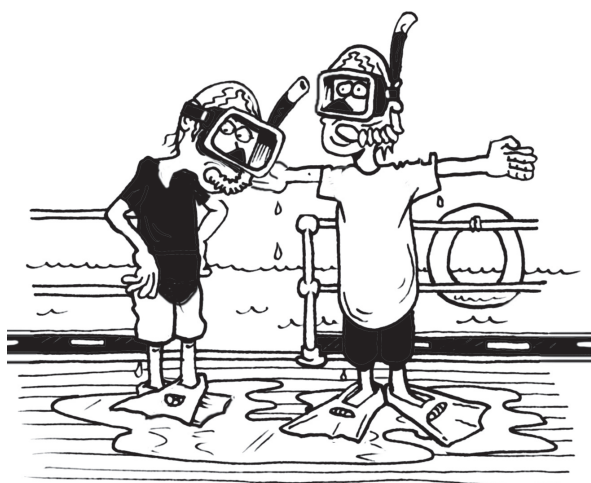
b Koh Samui?

c Phnom Penh?





- 4 Which months have the lowest average maximum temperature in
 - a Bangkok?
 - b Koh Samui?
 - c Phnom Penh?
- 5 What is the average minimum temperature in Phnom Penh during January?
- 6 What is the lowest average minimum temperature in Bangkok?
- 7 Which two months have the lowest average minimum temperature in Bangkok?
- 8 What is the average number of rain days in Koh Samui during January?
- 9 Which two months have the *highest* average number of rain days in Koh Samui?
- 10 Which two months have the *highest* average number of rain days in Bangkok?
- 11 Which three months have the *lowest* average number of rain days in Bangkok?
- 12 If Alicja and David visit Bangkok in September, can they expect rain on a majority of days?
- 13 Which three months have the *lowest* average minimum temperature in Bangkok?
- 14 Which three months have the *lowest* average maximum temperature in Phnom Penh?
- 15 Of the months with the lowest average maximum temperature in Phnom Penh, which month has the lowest average number of rainy days?
- 16 Which three months have the *highest* average number of rain days in Koh Samui?
- 17 Which three months have the *lowest* average number of rain days in Koh Samui?
- 18 Of the months with the lowest average number of rain days in Koh Samui, which month has the lowest average maximum temperature?
- 19 Alicja and David chose **February** for their holiday. Write a brief description of the weather (mention the average maximum temperature and the number of rain days) they should expect in
 - a Bangkok.
 - b Koh Samui.
 - c Phnom Penh.

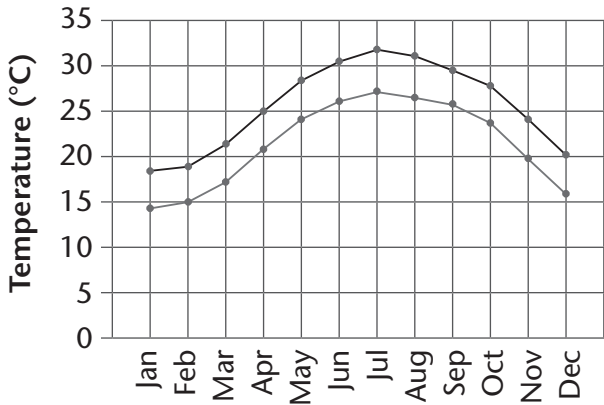




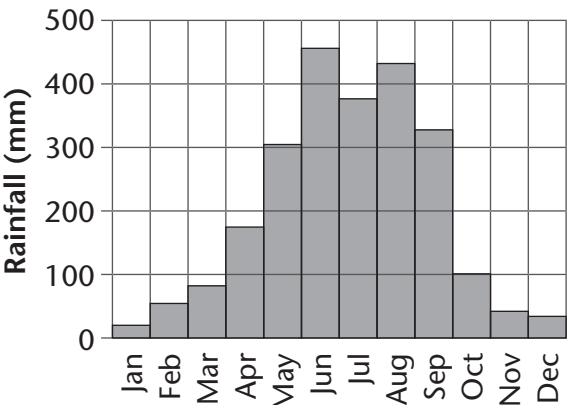
The graphs on this work sheet show the average maximum and minimum temperatures (in °C) and the average rainfall (in mm) for each month in three cities in eastern Asia.

HONG KONG

Average maximum and minimum temperature (°C)



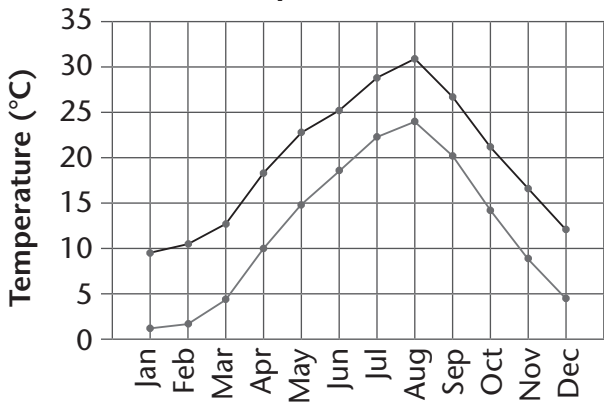
Average monthly rainfall (mm)



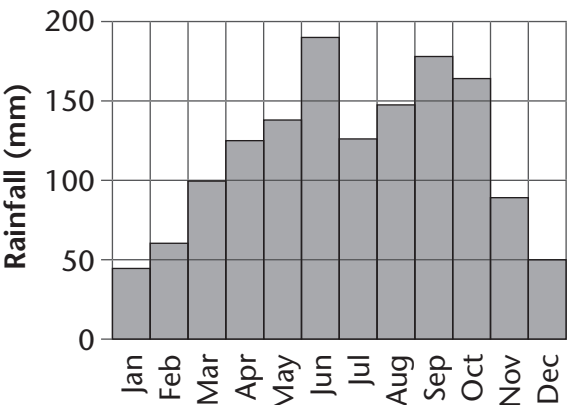
- 1 Which month has the highest average
 - a maximum temperature?
 - b minimum temperature?
 - c rainfall?
- 2 Which month has the lowest average
 - a maximum temperature?
 - b minimum temperature?
 - c rainfall?
- 3 Which three months have the *highest* average maximum temperature?

TOKYO

Average maximum and minimum temperature (°C)

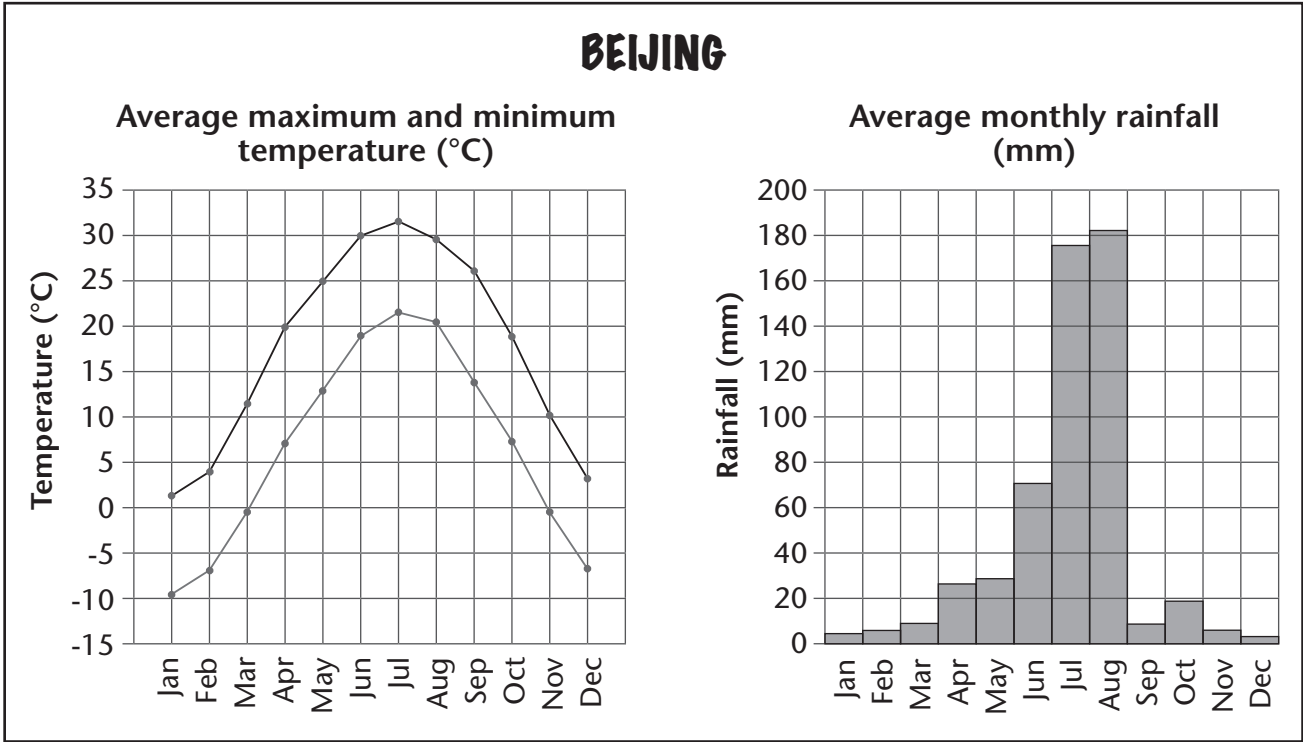


Average monthly rainfall (mm)





- 4 Which month has the highest average
 - a maximum temperature?
 - b rainfall?
- 5 Which month has the lowest average
 - a minimum temperature?
 - b rainfall?
- 6 Which three months have the *lowest* average rainfall?



- 7 Which month has the highest average
 - a maximum temperature?
 - b rainfall?
- 8 Which month has the lowest average
 - a minimum temperature?
 - b rainfall?
- 9 Which three months have the *highest* average rainfall?

Compare the graphs for Hong Kong, Tokyo and Beijing and answer these questions.

- 10 Which city has the highest average maximum temperature in January?
- 11 Which city has the highest average maximum temperature in May?
- 12 Which city has the lowest average minimum temperature in January?
- 13 Which city has the lowest average minimum temperature in September?
- 14 Which city has the highest average rainfall in March?
- 15 Which city has the lowest average rainfall in October?



The table below shows the mean (average) rainfall for each month at various cities in Australia, Pakistan, Malaysia and Papua New Guinea. The monthly averages have been rounded to the nearest 0.5 mm.

Mean monthly rainfall (in mm)

	Brisbane	Melbourne	Sydney	Islamabad (Pakistan)	Kuala Lumpur (Malaysia)	Port Moresby (Papua New Guinea)
Jan	133.0	47.5	101.5	56.0	163.0	192.0
Feb	142.0	48.0	118.0	73.5	144.5	140.5
Mar	96.5	50.5	130.0	90.0	218.5	190.0
Apr	67.0	57.5	126.5	62.0	285.0	105.0
May	64.5	56.0	121.0	39.0	184.0	56.0
Jun	55.5	49.0	130.5	62.0	127.0	21.5
Jul	22.5	47.5	98.5	267.0	129.0	14.0
Aug	45.0	50.0	80.5	310.0	145.5	12.0
Sep	32.0	58.0	69.0	98.0	192.0	14.5
Oct	87.0	66.5	77.5	29.5	272.5	15.0
Nov	103.5	60.5	84.0	18.0	275.5	40.0
Dec	159.5	59.5	78.0	37.5	230.5	98.0

Use the data in this table to answer the following questions.

- 1

Which city has the *highest* mean monthly rainfall during

a January?

b April?

c June?

d August?
- 2

Which city has the *lowest* mean monthly rainfall during

a January?

b May?

c July?

d November?
- 3

Which month has the *highest* mean rainfall in Melbourne?
- 4

Which month has the *highest* mean rainfall in Kuala Lumpur?
- 5

Which month has the *lowest* mean rainfall in Sydney?
- 6

Which month has the *lowest* mean rainfall in Port Moresby?



- 7 Which three months have the *highest* mean rainfall in Brisbane?
- 8 Which three months have the *lowest* mean rainfall in Kuala Lumpur?
- 9 Which three cities have the highest mean rainfall for December?
- 10 Which three cities have the lowest mean rainfall for May?
- 11 Calculate the average monthly rainfall in Kuala Lumpur, correct to the nearest millimetre.
- 12 Which months have a mean rainfall less than the monthly average in Kuala Lumpur?
- 13 In Sydney, the mean monthly rainfall for March is 130.2 mm.
On average, there are 14 rain days in March.
Calculate the average rainfall on these days, correct to 1 decimal place.



This table shows the average number of rain days each month in Brisbane and Port Moresby.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Brisbane	12	13	13	12	10	9	6	6	8	10	12	14
Port Moresby	18	16	17	11	9	6	4	4	5	6	7	12

- 14 In Brisbane, the mean monthly rainfall for August is 44.8 mm.
Calculate the average rainfall on each rain day for Brisbane in August.
Answer correct to 1 decimal place.
- 15 In Brisbane, December has the highest mean monthly rainfall (159.7 mm).
Calculate the average rainfall on each rain day for Brisbane in December.
Answer correct to 1 decimal place.
- 16 In Port Moresby, January has the highest mean monthly rainfall (192.2 mm).
Calculate the average rainfall on each rain day for Port Moresby in January.
Answer correct to 1 decimal place.
- 17 In Port Moresby, the mean monthly rainfall for March is 189.8 mm.
Calculate the average rainfall for each rain day for Port Moresby in March.
Answer correct to 1 decimal place.



The countries of East Asia shared more than 55% of Australia's world trade in 2011. This work sheet looks at Australia's trade with China, Indonesia, Japan, the Republic of Korea (South Korea), Singapore and Thailand. These countries were Australia's six main trading partners in East Asia in 2011. (The data for China does not include Hong Kong.)

Australia's exports and imports of goods and services (A\$ million)

	Exports			Imports		
Country	2009	2010	2011	2009	2010	2011
China	47 922	64 295	77 117	37 238	40 907	43 938
Indonesia	5 331	5 733	6 662	5 991	7 134	8 167
Japan	40 271	45 665	52 384	18 812	20 405	20 076
Republic of Korea	17 453	22 381	25 024	6 947	7 706	7 644
Singapore	8 277	7 443	9 526	14 642	14 167	18 157
Thailand	5 306	6 862	7 725	13 869	12 955	10 732
Total East Asia	145 220	176 055	204 392	118 315	126 550	131 882
Total World	249 602	284 211	313 255	254 428	268 994	294 933

Based on the DFAT publication: *Australia's trade with East Asia 2011*

1 What was the *total* value of Australia's **exports** to China for the 3 years 2009 to 2011?

Give your answers for questions 2 to 5 correct to the nearest million dollars.

- 2 What was the mean value of Australia's **exports** to China for the 3 years 2009 to 2011?
- 3 What was the mean value of Australia's **exports** to Indonesia for the 3 years 2009 to 2011?
- 4 What was the mean value of Australia's **imports** from China for the 3 years 2009 to 2011?
- 5 What was the mean value of Australia's **imports** from Japan for the 3 years 2009 to 2011?



- 6 What percentage of Australia's total **imports** (Total World), correct to 1 decimal place, were Australia's imports from East Asia
 - a in 2009?
 - b in 2011?
- 7 What percentage of Australia's total **exports** (Total World), correct to 1 decimal place, were Australia's exports to East Asia
 - a in 2009?
 - b in 2011?



- 8 What percentage of Australia's total exports to East Asia were Australia's exports to China in 2011?
Answer correct to one decimal place.
- 9 What percentage of Australia's total imports (Total World) were Australia's imports from Japan in 2011?
Answer correct to one decimal place.
- 10 With which of the six countries listed in the table were Australia's exports greater than imports in 2011?
- 11 With which of the six countries listed in the table were Australia's imports greater than exports in 2011?



Balance of trade = export value – import value
When imports are **greater** than exports, the balance of trade is **negative**.

- 12 Calculate the balance of trade with each country in 2011.
Record your answers in the first column of the following table.

Country	Balance of trade A\$ million	Total trade A\$ million	% share of world total
China			
Indonesia			
Japan			
Republic of Korea			
Singapore			
Thailand			

- 13 Calculate the mean balance of trade for these six countries.
Round your answer to the nearest million dollars.
- 14 Calculate the total trade (exports + imports) with each country in 2011.
Record your answers in the table in question 12.
- 15 Express the total trade as a percentage of Australia's total world trade (\$608 188 000 000).
Round the percentages to 2 decimal places and record your answers in the table in question 12.
- 16 Rank these countries in order of their percentage of Australia's total world trade.



In 2011, a report published by the Department of Foreign Affairs and Trade stated:

The value of total trade in goods and services to East Asia increased 11.1 per cent to \$336.3 billion in 2011 – since 2006, total trade values have increased by an average of 9.2 per cent per annum.

- *In comparison, Australia’s total trade with the world rose 9.9 per cent to \$608.2 billion.*
- *East Asia accounted for 55.3 per cent of Australia’s total trade in goods and services.*



Source: DFAT publication *Australia’s trade with East Asia 2011*

Matthew was reading an article he found on the internet while doing research for a school project. The article claimed:

Australia’s major merchandise exports to East Asia in 2011 were

- *Iron ore & concentrates – up 113.2 per cent to \$64.0 billion*
- *Coal – up 110.7 per cent to \$32.4 billion*
- *Crude petroleum – up 13.6 per cent to \$8.8 billion*
- *Gold – up 172.1 per cent to \$6.2 billion.*

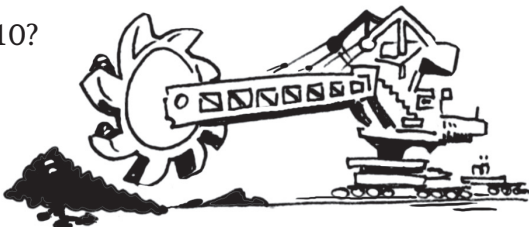
Jane did not think these statements were accurate and checked the data available from the Department of Foreign Affairs and Trade. She found the following data.

Australia’s top 6 merchandise exports to East Asia
(in A\$ million) (rank is based on 2011)

Commodity	2009	2010	2011	% growth 2010–11
Iron ore and concentrates	30 010	49 224	63 996	
Coal	29 598	29 266	32 386	
Crude petroleum	6 282	8 785	9 982	
Gold	2 248	3 578	6 156	
Wheat	1 927	2 599	4 096	57.6
Aluminium	3 830	4 061	4 015	-1.1

Based on the DFAT publication *Australia’s trade with East Asia 2011*

- 1 What was the value of exports of iron ore and concentrate in 2009?
Answer in A\$ billion. Note: \$1 billion = \$1000 million.
- 2 What was the value of exports of aluminium in 2010?
Answer in A\$ billion, correct to 1 decimal place.
- 3 What was the value of exports of wheat in 2011?
Answer in A\$ billion, correct to 1 decimal place.





- 4 What was the percentage growth of exports of wheat from 2010 to 2011?
- 5 In the table, the percentage growth of exports of aluminium is -1.1. Why is this negative?
- 6 Calculate the percentage growth in the value of the exports of Iron ore and concentrates, Coal, Crude petroleum and Gold from 2010 to 2011. Round each answer correct to 1 decimal place and record it in the space in the table.
- 7 Circle or highlight the errors in the statements Matthew copied from the internet article.
- 8 In your workbook, rewrite the statements using the correct data.
- 9 What is the mean percentage growth in the value of Australia's top 6 merchandise exports to East Asia from 2010 to 2011? Give your answer correct to 1 decimal place.

Jane also found the following data relating to Australia's imports.

Australia's top 6 merchandise imports from East Asia
(in A\$ million) (rank based on 2011)

Commodity	2009	2010	2011	% growth 2010–11
Refined petroleum	9 170	8 959	13 061	45.8
Crude petroleum	7 261	9 599	9 034	-5.9
Passenger motor vehicles	7 900	10 521	8 514	-19.1
Telecom equipment & parts	5 260	5 745	6 292	9.5
Computers	4 494	5 566	5 679	2.0
Goods vehicles	3 251	4 275	3 437	-19.6

Based on the DFAT publication *Australia's trade with East Asia 2011*

- 10 What was the value of imports of refined petroleum in 2009? Answer in A\$ billion.
- 11 What was the value of imports of computers in 2011? Answer in A\$ billion, correct to 1 decimal place.
- 12 Calculate the percentage growth in the value of the imports of goods vehicles from 2009 to 2010. Answer correct to 1 decimal place.
- 13 Calculate the percentage growth in the value of the imports of computers for the two year period 2009 to 2011. Answer correct to 1 decimal place.
- 14 What is the mean percentage growth in the value of Australia's top 6 merchandise imports from East Asia from 2010 to 2011? Give your answer correct to 2 decimal places.



Through the centuries the Chinese used characters for writing numbers. They were very advanced in their mathematics and it is believed that they had a highly developed number system as early as 1 500 BC.

There are characters representing the numbers 1 to 10. Other characters are used for larger numbers like 100 and 1 000.

The basic symbols are:

1	2	3	4	5	6	7	8	9	10	100	1 000
一	二	三	四	五	六	七	八	九	十	百	千
yī	èr	sān	sì	wǔ	liù	qī	bā	jiǔ	shí	bǎi	qiān

Writing numbers using the Chinese characters is similar to writing out numbers in English (e.g. 'one thousand, nine hundred, forty-five'). Since it reflects spoken language, the Chinese system does not use the positional system we use with Hindu-Arabic numerals.

Multiples of 10, 100 and 1 000 are constructed using a multiplication principle.

For example, 30 is written vertically, using the symbols for 3 and 10 as 3 times 10: 三十

- 1 The following characters represent multiples of 10 or 100. Write the Hindu-Arabic numerals for each number.

a 二十	b 五十	c 七十	d 四百	e 九百
十	十	十	百	百
f 三百	g 六十	h 五百	i 八百	j 四十
百	十	百	百	十

For the numbers 11 to 19, the leading one (一) indicating 1×10 is usually omitted.

For example, 15 is usually written as 10 + 5: 十五

- 2 Write the Hindu-Arabic numerals for these numbers.

a 十二	b 十四	c 十六	d 十七	e 十八
十	十	十	十	十
二	四	六	七	八



Traditionally, Chinese numerals are written down the page, from top to bottom.

The symbols are grouped in pairs. Each pair of numbers is multiplied to give the number of tens, hundreds, thousands and so on. These products are added to give the number.

Examples:

$$\begin{array}{r} 1 \quad \text{六} \quad 6 \\ \quad \times \\ \text{十} \quad 10 \\ \quad + \\ \text{八} \quad 8 \end{array}$$

Answer: 68

$$\begin{array}{r} 2 \quad \text{三} \quad 3 \\ \quad \times \\ \text{百} \quad 100 \\ \quad + \\ \text{五} \quad 5 \\ \quad \times \\ \text{十} \quad 10 \\ \quad + \\ \text{二} \quad 2 \end{array}$$

Answer: 352

3 What numbers are represented by these Chinese numerals?

a 二十五

b 三十七

c 四十六

d 七十八

e 九十四

f 百七十五

g 二百五十六

h 四百三十八

i 五百二十四

j 六百九

k 二千七百六十九

l 四千三百十六

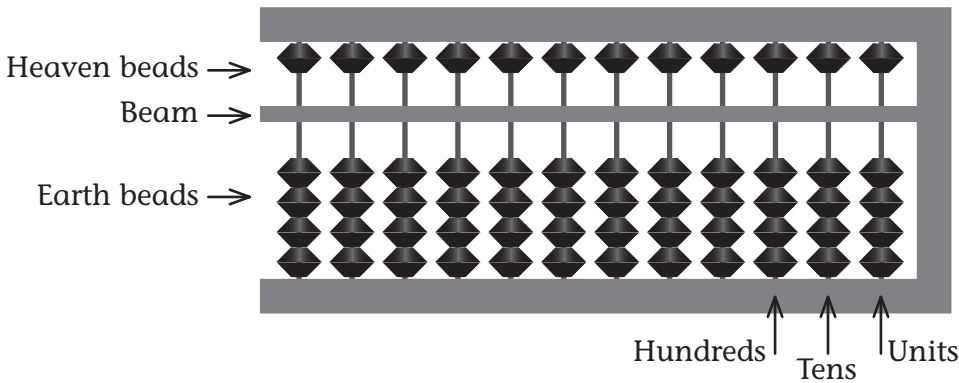
m 千八百三十一

n 六千四十二

o 九千三百五



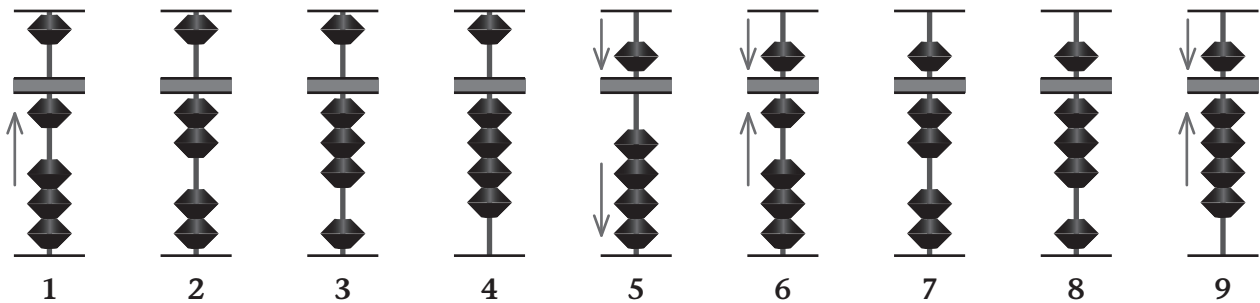
The abacus was one of the earliest calculating machines developed. It is used by pushing beads up or down. The diagram shows a type of abacus used in Japan.



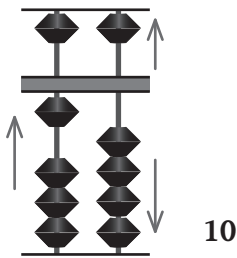
The value of the bead is determined by the column it lies in. The place values work right to left. The far right column of beads is the units column. Moving to the left, we have the tens column then the hundreds column and then the thousands column.

Above the beam the corresponding values of the heaven beads, right to left, are 5, 50, 500, 5 000 and so on.

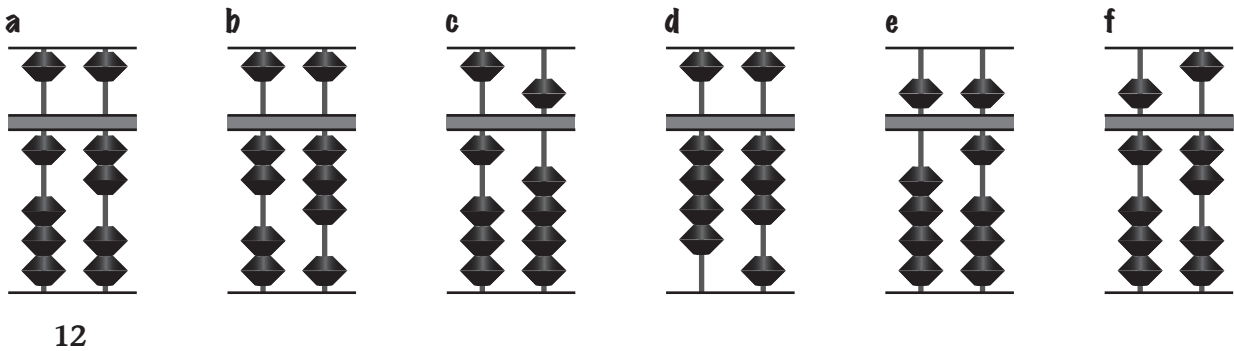
These diagrams show the positions of the beads for the digits 1 to 9.



10 is shown with one raised bead in the tens column. The units column is reset to zero.

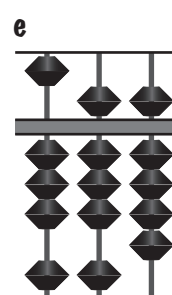
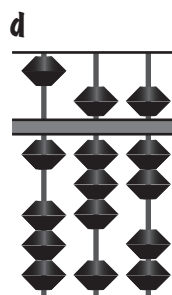
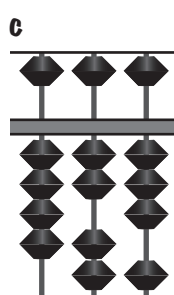
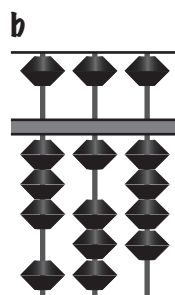
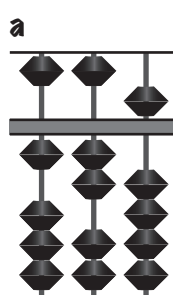


1 What two-digit numbers are represented in these diagrams? The first answer is shown as an example.

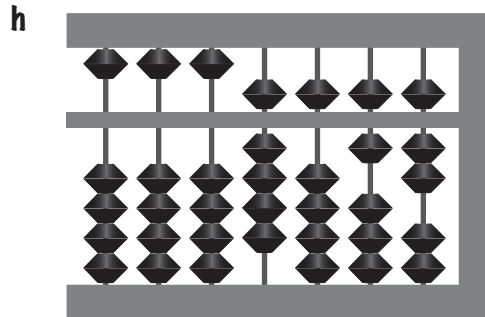
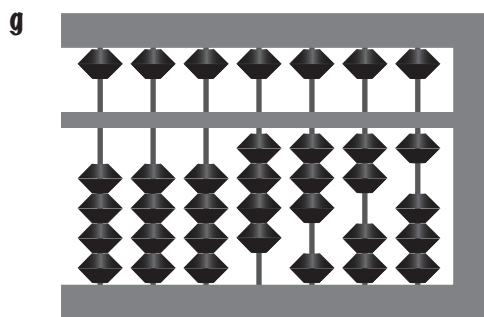
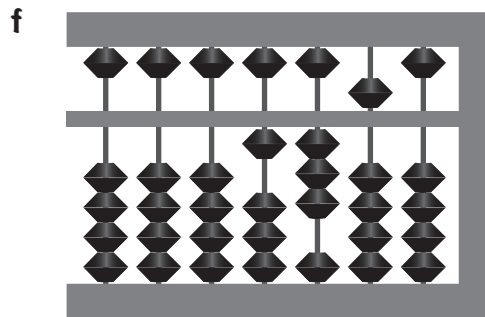
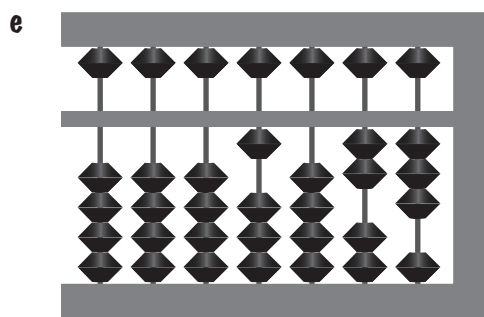
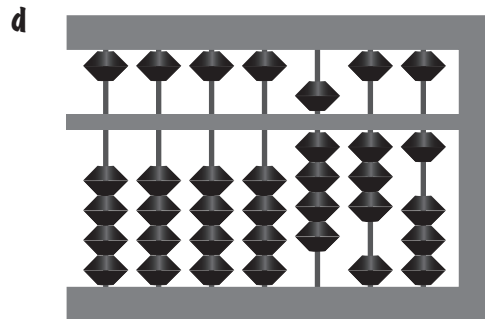
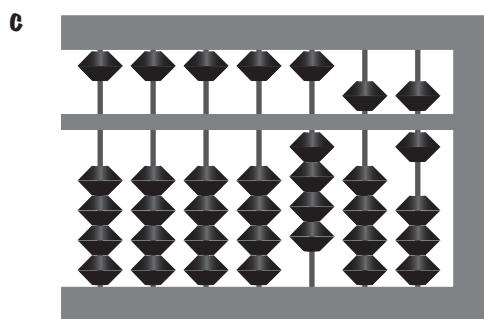
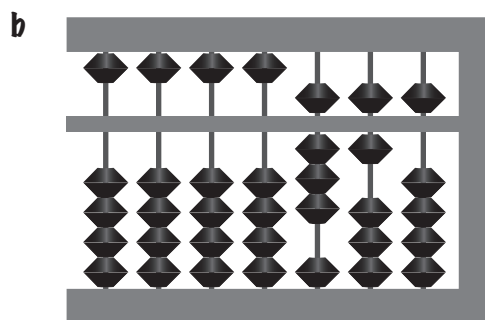
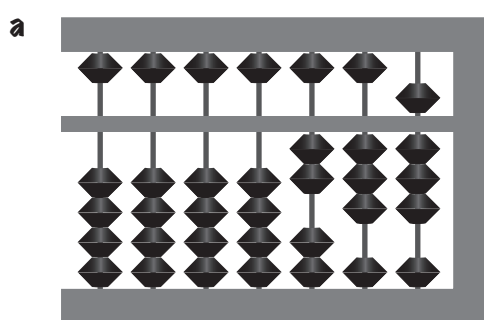




2 What three-digit numbers are represented in these diagrams?



3 What numbers are represented in these diagrams?





1 Time lines in Asian history

- 1 a 1276 b 1644 c 1842 d 1949
- 2 B
- 3 a 1368 b 1644 c 276
- 4 268
- 5 a 92 b 368 c 636
- 6 155
- 7 a 1526 b 1773 c 1950
- 8 A
- 9 a 103 b 177

2 Birds of New Guinea

- 1 $\frac{35}{44}$
- 2 $\frac{50}{430} = \frac{5}{43}$
- 3 $\frac{44}{110} = \frac{2}{5}$
- 4 13.6%
- 5 a $\frac{1060}{2000} = \frac{53}{100}$ b 53%
- 6 58 cm
- 7 $\frac{30}{36} = \frac{5}{6}$
- 8 28.0%
- 9 100 g
- 10 $\frac{135}{235} = \frac{27}{47}$
- 11 174.1%
- 12 63.8%
- 13 22 cm
- 14 26 cm
- 15 34 cm

3 Elephants in Thailand

- 1 a 3% b 22.4% c 60% d 150%
- 2 a 60% b 1.4%
- 3 2.8%
- 4 a 16 b 20% c 64%
- 5 69.4%
- 6 a 1.6 b 480
- 7 a 1 300 kg b 27.1%
- 8 a 39% b 2.8%
- 9 0.15%

4 Bullet trains in Japan

- 1 2 h 39 min
- 2 a 2 h 57 min b 2.95 h
- 3 5:21 pm
- 4 a 2.3 h b 223.0 km/h
- 5 a 1.42 h b 156.3 km/h
- 6 a 3.17 h b 136.9 km/h
- 7 a 1.05 h b 147 km
- 8 a 4.15 h b 894 km
- 9 2 h 34 min
- 10 1 h 36 min
- 11 a 3 h 30 min b 2:05 pm

5 Local markets

- 1 a bananas: 15 000 VND, oranges: 19 200 VND, rambutans: 12 500 VND

b 46 700 VND c 3 300 VND

- 2 a mangoes: 27 000 VND, lychees: 35 000 VND, apples: 12 800 VND
b 74 800 VND c 5 200 VND
- 3 37 200 VND
- 4 a \$1.19 b \$1.90 c \$2.86 d \$3.57 e \$4.76
f \$6.19 g \$8.10 h \$11.90
- 5 \$2.22
- 6 \$3.56
- 7 \$1.77
- 8 a 81 000 VND b \$3.86
- 9 \$3.10

6 Tall buildings

2	Petronas Twin Towers	450 m
	Taipei 101	510 m
	Eureka Tower	300 m
	Q1	320 m

- 3 a 2:3 b 10:17 c 16:21 d 32:45 e 32:51
f 7:5 g 3:2 h 17:14
- 4 $\frac{2}{3}$
- 5 $\frac{3}{2} = 1\frac{1}{2}$
- 6 $\frac{15}{16}$
- 7 a 2:7 b 7:2 c $3\frac{1}{2}$

7 Population growth

- 1 17 096 000 (or 17.096 million)
- 2 1 053 898 000
- 3 Australia, Thailand, Philippines, Indonesia, India
- 4 a 1 022 000 b 5.98% c 1.2%

5	Australia	1.2%	1.2%	1.3%	1.8%
	India	3.0%	1.9%	1.6%	1.5%
	Indonesia	1.6%	1.4%	1.3%	1.1%
	Philippines	2.5%	2.3%	2.1%	1.8%
	Thailand	0.9%	1.2%	1.1%	0.7%

- 6 Australia, India, Indonesia, Philippines
- 7 Thailand
- 8 Philippines
- 9 Australia
- 10 Philippines
- 11 Thailand
- 12 Australia
- 13 a 305 100 b 27 186 400 c 4 365 700

8 Population ratios

- 1 females
- 2 India
- 3 Cambodia
- 4 China
- 5 Cambodia
- 6 99:100
- 7 a 100:100 b 1:1
- 8 a 96:100 b 24:25
- 9 19:20
- 10 27:25



- 11 Malaysia
- 12 Cambodia
- 13 92 173 000
- 14 11 190 000
- 15 Cambodia
- 16 China, India
- 17 China
- 18 Cambodia, Indonesia
- 19 105:100 = 21:20
- 20 115:100 = 23:20
- 21 104:100 = 26:25
- 22 Australia
- 23 China

9 Islamic design

- 1 b 2 c 2
- 2 b 4 c 4
- 3 a 6 b 6 c 0 d 6

10 Temples of Angkor

- 1 5.6 km
- 2 3 650 m or 3.65 km
- 3 820 000 m² or 0.82 km²
- 4 a 1.95 km² b 1 950 000 m²
- 5 1 120 m × 920 m
- 6 919 600 m² or 0.9196 km²
- 7 4 512.5 m²
- 8 24 m
- 9 a 2 400 m or 2.4 km b 350 000 m²
- 10 800 m
- 11 a 560 000 m² b 700 m
- 12 a 3 km b 3.2 km c 9 km² d 10.24 km² e 1.24 km²

11 The Great Wall of China

- 1 a 70.7% b 4.1% c 25.2%
- 2 a 10 100 m² b 46 460 m³
- 3 9 586 m³
- 4 a 3 900 m² b 39 000 m³
- 5 a 35.75 m² b 357.5 m³ c 178 750 m³
- 6 2 655 km

12 Australian time zones and Asian countries

- 1 a 8:00 am b 7:30 am c 6:00 am d 8:00 am
- 2 a 4:30 pm b 4:00 pm
- 3 a 12:30 pm b 10:30 am c 12 noon d 12 noon
- 4 a 1:30 pm b 2:00 pm
- 5 a 5:00 pm b 4:30 pm c 3:00 pm d 3:00 pm
- 6 a 11:30 am b 1:30 pm
- 7 8:30 am
- 8 8:00 pm
- 9 a 10:30 am b 8:00 pm
- 10 a 14:30 b 14:30
- 11 a 12:45 b 11:15
- 12 a 22:10 b 21:40
- 13 17:25
- 14 13:30
- 15 23:40
- 16 08:45 (next day)

13 International time zones

- 1 11:30 am
- 2 a 9:00 am b 9:00 am c 7:00 am d 8:30 am
- 3 7:00 am
- 4 a 6:00 pm b 6:00 pm c 5:30 pm d 4:00 pm
- 5 11:00 am
- 6 1:30 pm
- 7 a 2:00 pm b 4:00 pm c 4:00 pm d 3:30 pm
- 8 1:00 pm
- 9 12:30 pm
- 10 a 3:00 am b 3:30 am c 3:30 am d 1:30 am
- 11 a 5:00 pm b 12 noon c 10:30 am
- 12 10:30 pm
- 13 a 1:00 pm b 10:30 pm
- 14 17:50
- 15 19:35
- 16 4 h 15 min

14 Forests in South East Asia

- 1 10 094 000
- 2 7 665 000
- 3 52%
- 4 Philippines, Vietnam
- 5 Philippines, Thailand, Vietnam
- 6 Indonesia
- 7 Cambodia
- 8 a 322 000 b 69 000
- 9 a 6 726 000 b 3 986 000
- 10 a 35.5% b 21.0%
- 11 a 0.6% b 25.5%

15 How hot? How cold?

- 1 32°C
- 2 a 35°C b 33°C c 35°C
- 3 a 31°C b 28°C c 30°C
- 4 a December b January c November and December
- 5 22°C
- 6 21°C
- 7 January, December
- 8 10
- 9 October, November
- 10 August, September
- 11 January, February, December
- 12 Yes
- 13 January, February, December
- 14 October, November, December
- 15 December
- 16 September, October, November
- 17 February, March, April
- 18 February
- 19 a average max 32°C; 1 rain day
b average max 30°C; 6 rain days
c average max 33°C; 2 rain days

16 Climate graphs

- 1 a July b July c June
- 2 a January b January c January

Selected answers (cont'd)



- 3 June, July, August
- 4 a August b June
- 5 a January b January
- 6 January, February, December
- 7 a July b August
- 8 a January b December
- 9 June, July, August
- 10 Hong Kong
- 11 Hong Kong
- 12 Beijing
- 13 Beijing
- 14 Tokyo
- 15 Beijing

17 Annual rainfall

- 1 a Port Moresby b Kuala Lumpur c Sydney d Islamabad
- 2 a Melbourne b Islamabad c Port Moresby d Islamabad
- 3 October
- 4 April
- 5 September
- 6 August
- 7 January, February, December
- 8 February, June, July
- 9 Kuala Lumpur, Brisbane, Port Moresby
- 10 Islamabad, Melbourne, Port Moresby
- 11 197 mm
- 12 January, February, May, June, July, August, September
- 13 9.3 mm
- 14 7.5 mm
- 15 11.4 mm
- 16 10.7 mm
- 17 11.2 mm

18 Australia's trade with East Asia

- 1 \$189 334 000 000 (or \$189 334 million)
- 2 \$63 111 000 000 (or \$63 111 million)
- 3 \$5 909 000 000
- 4 \$40 694 000 000
- 5 \$19 764 000 000
- 6 a 46.5% b 44.7%
- 7 a 58.2% b 65.2%
- 8 37.7%
- 9 6.8%
- 10 China, Japan, Republic of Korea
- 11 Indonesia, Singapore, Thailand

Country	Balance of trade A\$ million
China	33 179
Indonesia	-1 505
Japan	32 308
Republic of Korea	17 380
Singapore	-8 631
Thailand	-3 007

- 13 \$11 621 000 000 (or \$11 621 million)

14-15

Country	Total trade A\$ million	% share of world total
China	121 055	19.90
Indonesia	14 829	2.44
Japan	72 460	11.91
Republic of Korea	32 668	5.37
Singapore	27 683	4.55
Thailand	18 457	3.03

- 16 China, Japan, Republic of Korea, Singapore, Thailand, Indonesia

19 Growth of imports and exports

- 1 \$30.01 billion
- 2 \$4.1 billion
- 3 \$4.1 billion
- 4 57.6%
- 5 There was a decrease.

6	Iron ore and concentrates	30.0
	Coal	10.7
	Crude petroleum	13.6
	Gold	72.1

- 7 Iron ore & concentrates: up 113.2 per cent to \$64.0 billion
Coal: up 110.7 per cent to \$32.4 billion.
Crude petroleum: up 13.6 per cent to \$8.8 billion
Gold: up 172.1 per cent to \$6.2 billion
- 8 Iron ore & concentrates: up 30 per cent to \$64.0 billion
Coal: up 10.7 per cent to \$32.4 billion.
Crude petroleum: up 13.6 per cent to \$10 billion
Gold: up 72.1 per cent to \$6.2 billion
- 9 30.5%
- 10 \$9.17 billion
- 11 \$5.7 billion
- 12 31.5%
- 13 26.4%
- 14 2.12%

20 Chinese numerals

- 1 a 20 b 50 c 70 d 400 e 900 f 300 g 60 h 500 i 800 j 40
- 2 a 12 b 14 c 16 d 17 e 18
- 3 a 25 b 37 c 46 d 78 e 94 f 175 g 256 h 438 i 524 j 609 k 2 769 l 4 316 m 1 831 n 6 042 o 9 305

21 The abacus

- 1 a 12 b 23 c 15 d 43 e 56 f 62
- 2 a 125 b 314 c 423 d 187 e 389
- 3 a 238 b 865 c 456 d 931 e 1 023 f 1 350 g 4 321 h 9 567

Acknowledgements

pp16-18: United Nations ESCAP, 2011, *Statistical Yearbook for Asia and the Pacific 2011*, viewed 7 August 2013, <<http://www.unescap.org/stat/data/syb2011/>>

pp30-31: Food and Agriculture Organization of the United Nations, 2011, *Southeast Asia Subregional Report, Asia-Pacific Forestry Sector Outlook Study II*, <<http://www.fao.org/docrep/013/i1964e/i1964e00.pdf>>. Reproduced with permission.

pp38, 40-41: Department of Foreign Affairs and Trade, 2012, *Australia's trade with East Asia 2011*, viewed 4 June 2013, <<http://www.dfat.gov.au/publications/stats-pubs/Australia-trade-with-east-asia-2011.pdf>>

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